Certified World Models: Predictability Across Configuration, Horizon, and Resolution

Scale buys interpolation; structure buys a certificate.

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Abstract

Scaling a world model lowers its average error on the data manifold but issues **no guarantee about any specific unseen situation**. We give a different *kind* of result for **equivariant** latent world models — a **predictability certificate**: a provable, computable region of situations the model is guaranteed to handle *without further training or data*. The certificate spans three orthogonal axes at once: **configuration** (the exponentially large monoid $\langle S \rangle$ generated by k primitive symmetries, certified by checking the k generators), **horizon** T (how far ahead the guarantee reaches), and **resolution** ϵ (at what level of detail). A master theorem makes this precise: under exact equivariance with an orthogonal latent representation the whole-pipeline error is *invariant* over all of $\langle S \rangle$ (Theorem A), and a spectral refinement (Theorem B) shows that each latent channel is certified to a horizon $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$ set by its Lyapunov exponent — so the certified region is the **coarse-invariant, slow, low-composition** corner. We confirm all three axes at small scale: a model trained on the 6 generators of $\mathbb{Z}_2^6$ certifies all 2^6=64 compositions to machine precision while a matched non-equivariant baseline degrades monotonically; a learned predictor recovers a planted Lyapunov exponent to within 0.4% and its certified horizon obeys the $\log(1/\epsilon)$ law — and, on a learned model of the genuinely chaotic **Lorenz** system, the learned model's Lyapunov exponent (the certified-horizon slope) **matches the true** exponent to 1–8%, with a scope theorem (Proposition 7) characterizing when this lift holds (spectrally non-degenerate dynamics) and explaining where it does not (near-neutral dynamics); on an SO(2) mechanical system the conserved (slowest) quantities organize into the architecturally invariant channels — a **Noether hinge** linking the configuration axis to the horizon axis, whose forward direction (*conserved* ⇒ *slow*) we prove: a conserved charge's prediction error grows linearly, not exponentially, in its conservation defect; and a non-equivariant network scaled across an 88× parameter range buys in-distribution interpolation (even beating the equivariant model there) yet over the unseen orbit stays 10–155× above the equivariant floor and never reaches it. Finally, on **real PushT contact dynamics** — physics simulated by an engine we did not author — a learned equivariant world model's multi-step rollout error is *exactly* flat over the orbit and competitive in-distribution, while no non-equivariant baseline across a 160× parameter ladder (up to 16× the equivariant model's size) reaches its out-of-distribution floor; and run through a G-equivariant planner, the same certificate becomes *task* competence — closed-loop pose control whose error is orbit-invariant to the float floor on every seed, where a scaled baseline degrades. *Scale buys interpolation; structure buys a certificate.* The result is a single, **runnable** criterion (Algorithm 1, ≤ 20 lines) for *what* an equivariant world model can certifiably predict, and a structural reading of *why* celestial mechanics is forecastable for millennia while weather is not.

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1. Introduction

Two debates organize modern world-model research. The first is *generation vs. abstraction*: pixel-space simulators (diffusion and video models) against latent predictive models that predict representations rather than pixels. The second is *scale vs. structure*: the view that brute-force scaling eventually wins, against the geometric-learning view that symmetry priors buy sample efficiency. Both are usually run as **degree** contests — who needs fewer pixels, who needs less data.

We argue the more useful question is a **kind** contest. Scaling produces a model with low *average* error on the data distribution, but it cannot certify anything about a *named, specific* situation the data did not cover. Structure can. For an equivariant world model we can write down, in advance and without further data, the exact set of situations on which the model's behaviour is **guaranteed** — a *predictability certificate*. This reframes "is structure worth it?" from a benchmark race into a question about *what kind of statement each approach can make*.

The certificate has three orthogonal axes:

- **Configuration** $w \in \langle S \rangle$: the model generalizes across the exponentially large monoid generated by k primitive symmetries $S = \{g_1, \ldots, g_k\}$, and we *certify the whole monoid by checking only the k generators*.
- **Horizon** T: how far into the future the guarantee reaches.
- **Resolution** ϵ: at what level of detail the guarantee holds.

Our contributions are:

1. **A three-axis master theorem** (§3): composition closure (k checks $\Rightarrow$ an exponential set), an exact certificate under exact equivariance (Theorem A) together with its **converse** (Lemma 2: the certificate is *equivalent* to equivariance, so no non-equivariant model has it at any size), and a spectral degradation law $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$ (Theorem B) — **tight**, with a matching lower bound (Proposition 6: *approximate* equivariance is horizon-limited, so only exact structure or conservation reaches an unbounded horizon) and a **scope characterization** (Proposition 7: the local-spectrum horizon governs a *learned* model on spectrally non-degenerate dynamics, $\lambda_1 > 0$, and is vacuous on near-neutral dynamics; and **Proposition 8**: the learned model *recovers* the chaos rate because the certified horizon is finite and finite-horizon Lyapunov exponents are C^1-continuous — a $O(\delta)$ model-fidelity bias, not the shadowing-transfer the asymptotic exponent forbids) — whose certified region is the coarse-invariant-slow-low-$|w|$ corner, with a **quantitative scale-vs-structure separation** (§3.3: structure certifies the whole ϵ-independent orbit; the best L-Lipschitz non-equivariant learner certifies only an ϵ/L-tube around its data). Figure 1 is the whole picture at a glance.
2. **The Noether hinge** (§4): the bridge — that the group-invariant/equivariant channels are the dynamically slow (long-horizon-certifiable) ones — linking the configuration axis to the horizon axis. A representation-theoretic **placement principle** (Proposition 4) proves *which* isotypic block must carry each conserved charge and why 3D angular momentum is recoverable only at a unique degree-2 cross product; **Proposition 5** proves the *forward* direction (*conserved* $\Rightarrow$ *slow*: a charge conserved to one-step defect η has prediction error $\leq T\eta$ — linear, never the exponential $e^{\lambda T}$ of a chaotic channel — so it is certified to all horizons at η=0). What remains measured is the dynamical-symmetry hypothesis and the size of η.
3. **Empirical confirmation of all three axes at small scale** (§5), including the headline contrast *scale buys interpolation; structure buys a certificate*, a keystone validation on **real physics-engine contact dynamics** (PushT) where the certificate holds for a *learned* model of dynamics we did not design, and a **closed-loop** instantiation where, run through a planner, the certificate becomes *task* competence — orbit-invariant pose control where a scaled baseline degrades — a lift from the circle to the **non-abelian** SO(3) on 3D point clouds (the certificate is not SO(2)-specific), a lift to raw **pixels** (Experiment 13) where *frame averaging* makes the exact certificate **accuracy-neutral** — an equivariant pixel model matches an unconstrained CNN and is horizon-stable, so the prior costs nothing — and a lift of the **horizon law to a learned model of genuinely chaotic dynamics** (Lorenz, Experiment 14: the learned model's Lyapunov exponent, read as the certified-horizon staircase slope, *matches* the true exponent to 1–8%, instantiating Proposition 7(a)).

We are explicit about scope (§7): this is a mechanism-and-theory paper with 1–2-GPU proof-of-principle, not a scaled benchmark, and the hinge's lift to *approximate* symmetry is open.

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2. Setup

An encoder $E : \mathscr{X} \to \mathscr{Z}$ maps states to latents and an action-conditioned predictor $f : \mathscr{Z} \times \mathscr{A} \to \mathscr{Z}$ advances them; the true environment transition is $\Phi : \mathscr{X} \times \mathscr{A} \to \mathscr{X}$. A finite generating set $S = \{g_1, \ldots, g_k\}$ generates a monoid $\langle S \rangle$ whose elements — words $w = g_{i_1} \cdots g_{i_m}$ of length $|w| = m$ — act on states by $g \cdot x$, on latents by a representation $\rho(g)$, and on actions by $\sigma(g)$. Write $f^{(T)}(z; \vec{a})$ for the T-step latent rollout under actions $\vec{a} = (a_1, \ldots, a_T)$, x_T^{true} for the T-step true state, and

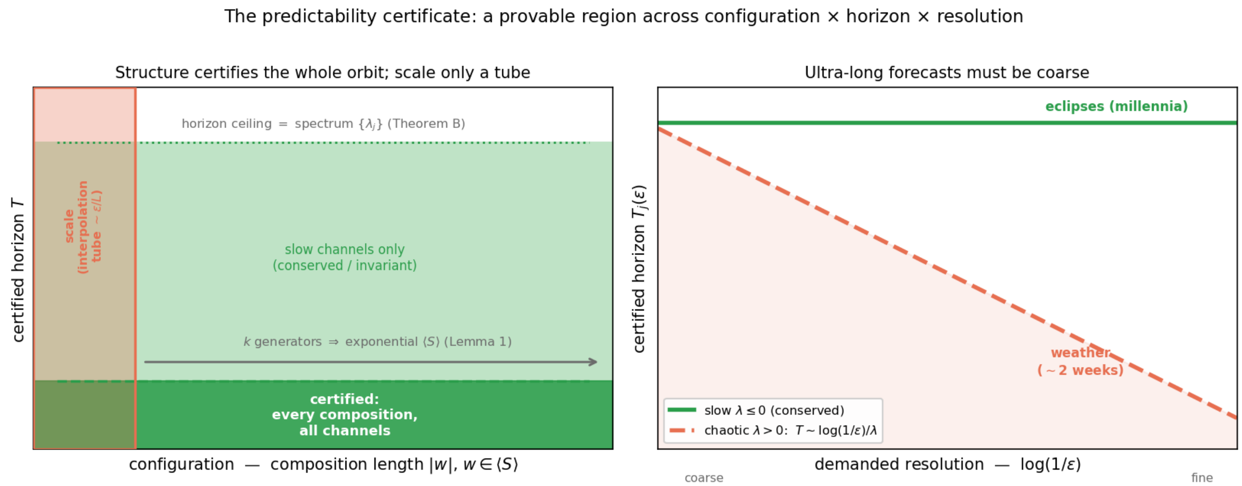

Figure 1: The predictability certificate at a glance. **Left:** in the configuration × horizon plane, an equivariant model certifies the *entire* generated monoid $\langle S\rangle$ — every composition, from k generator checks (Lemma 1) — up to a horizon ceiling set by the predictor spectrum $\{\lambda_j\}$ (Theorem B); a non-equivariant model of any size certifies only a small interpolation *tube* around its training set ($\sim \epsilon/L$, §3.3). **Right:** the horizon × resolution trade-off $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$ — conserved/invariant (slow, $\lambda \leq 0$) channels are certified to all horizons (eclipses, millennia), chaotic ($\lambda > 0$) channels shrink as the demanded resolution sharpens (weather, ~two weeks). *Scale buys interpolation; structure buys a certificate.*

$$\mathrm{Err}_T(x;\vec{a}) = \|f^{(T)}(E(x);\vec{a}) - E(x_T^{\mathrm{true}})\|$$

for the whole-pipeline prediction error, where $\|\cdot\|$ is the Euclidean norm on $\mathscr{Z}$ (the experiments use relMSE, its square). We use the following assumptions, **each checkable on the k generators**:

- **(A1) Encoder equivariance:** $E(g_i \cdot x) = \rho(g_i)\,E(x)$.
- **(A2) Predictor equivariance:** $f(\rho(g_i)z,\ \sigma(g_i)a) = \rho(g_i)\,f(z,a)$.
- **(A3) The group is a symmetry of the *dynamics*:** $\Phi(g_i \cdot x,\ \sigma(g_i)a) = g_i \cdot \Phi(x,a)$, so that rolling out the *transformed* state equals transforming the *rolled-out* state, $(g_i \cdot x)_T^{\mathrm{true}} = g_i \cdot x_T^{\mathrm{true}}$ under $\sigma(g_i)\vec{a}$.
- **(A4) $\rho(g_i)$ orthogonal:** $\rho(g_i)^\top \rho(g_i) = I$ (norm-preserving on $\mathscr{Z}$).
- **(A5) Equivariant planner and invariant cost (closed-loop clause only):** the planner satisfies $\pi(\rho(g_i)z, \rho(g_i)z^\star) = \sigma(g_i)\,\pi(z,z^\star)$ and the cost is ρ-invariant, $J(\rho(g_i)z; \rho(g_i)z^\star) = J(z;z^\star)$. This is needed *only* for the closed-loop J identity, not for the prediction-error identity. §5.6 instantiates such a planner in latent space; §5.9 instantiates one on **real PushT contact dynamics** — an equivariant CEM-MPC (isotropic exploration covariance, a rotation-invariant disk action bound, and scene-covariant action noise) — and confirms the resulting closed-loop task error is exactly orbit-invariant.

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3. The Predictability Certificate

3.1 The configuration axis (Theorem A)

Lemma 1 (composition closure). If (A1)–(A4) hold on each generator $g_i \in S$, they hold on every word $w \in \langle S\rangle$, and ρ restricted to $\langle S\rangle$ is an orthogonal-matrix *monoid homomorphism*.

Proof. Define $\rho(w) := \rho(g_{i_1})\cdots\rho(g_{i_m})$. For (A2), induct on m: $f(\rho(w)z, \sigma(w)a) = f(\rho(g_{i_1})[\rho(g_{i_2}\cdots)z],\ \sigma(g_{i_1})[\sigma(g_{i_2}\cdots)a]) = \rho(g_{i_1})f(\rho(g_{i_2}\cdots)z, \sigma(g_{i_2}\cdots)a) = \rho(w)f(z,a)$, applying the single-generator (A2) once per step. (A1) and (A3) lift identically by composition; (A4) lifts because a product of orthogonal matrices is orthogonal. □

Hence k **generator checks certify the entire (exponentially large) generated monoid.**

Theorem A (exact configuration certificate). Under (A1)–(A4), for every $w \in \langle S \rangle$, every horizon T, and every action sequence $\vec{a}$,

$$\mathrm{Err}_T(w \cdot x;\ \sigma(w)\vec{a}) = \mathrm{Err}_T(x; \vec{a}), \qquad \text{and, for an equivariant planner,} \quad J(w \cdot x;\ w \cdot \mathrm{goal}) = J(x; \mathrm{goal}).$$

Proof. **(i) Rollout equivariance.** By induction on T: $f^{(1)} = f$ is equivariant by (A2) and Lemma 1; assuming $f^{(T)}(\rho(w)z; \sigma(w)\vec{a}) = \rho(w) f^{(T)}(z; \vec{a})$,

$$f^{(T+1)}(\rho(w)z; \sigma(w)\vec{a}) = f\big(f^{(T)}(\rho(w)z; \sigma(w)\vec{a}),\ \sigma(w)a_{T+1}\big) = f\big(\rho(w)f^{(T)}(z; \vec{a}),\ \sigma(w)a_{T+1}\big) = \rho(w)f^{(T+1)}(z; \vec{a}).$$

(ii) Predicted latent at $w \cdot x$**.** $f^{(T)}(E(w \cdot x); \sigma(w)\vec{a}) \overset{\text{(A1)}}{=} f^{(T)}(\rho(w)E(x); \sigma(w)\vec{a}) \overset{\text{(i)}}{=} \rho(w) f^{(T)}(E(x); \vec{a})$. **(iii) Target at** $w \cdot x$**.** $E\big((w \cdot x)^{\mathrm{true}}_T\big) \overset{\text{(A3)}}{=} E\big(w \cdot x^{\mathrm{true}}_T\big) \overset{\text{(A1)}}{=} \rho(w)\, E(x^{\mathrm{true}}_T)$. **(iv) Cancellation.** Subtracting (iii) from (ii) and using (A4) (an orthogonal ρ preserves the Euclidean norm),

$$\mathrm{Err}_T(w \cdot x; \sigma(w)\vec{a}) = \|\rho(w)[f^{(T)}(E(x); \vec{a}) - E(x^{\mathrm{true}}_T)]\| = \|f^{(T)}(E(x); \vec{a}) - E(x^{\mathrm{true}}_T)\| = \mathrm{Err}_T(x; \vec{a}).$$

The closed-loop identity is the same computation under (A5): the equivariant planner (lifted to words by Lemma 1) selects the $\sigma(w)$-image of the same actions, and the ρ-invariant cost is unchanged. □

Theorem A requires **(A3)** — the group must be a symmetry of the *dynamics*, not merely of the encoder. Where the dynamics break the symmetry, the certificate degrades by the amount of that break (Theorem B). The error is held *constant across the orbit*, not made small: $\mathrm{Err}_T(x)$ itself can be large, and we report absolute errors alongside ratios throughout (§7).

Theorem A is sufficient; the next lemma shows equivariance is also *necessary* for the guarantee, so the certificate is not merely a consequence of equivariance but is **equivalent** to it — which is what makes the impossibility for non-equivariant models a theorem rather than a slogan.

Lemma 2 (the certificate characterizes equivariance). Let $\rho : G \to O(\mathscr{Z})$ act *freely* on an open $U \subseteq \mathscr{Z}$, and let $\mathscr{D}_G$ be the equivariant maps $\mathscr{Z} \to \mathscr{Z}$. If a predictor f's one-step error $\|f - \Phi\|$ is orbit-constant on U for **every** target $\Phi \in \mathscr{D}_G$, then f is equivariant on U. *Proof.* Fix $z \in U$, g with $\rho(g)z \in U$. For any $c \in \mathscr{Z}$, freeness makes $\Phi(\rho(h)z) := \rho(h)c$ well-defined on the orbit of z (extend by 0 elsewhere; both pieces lie in $\mathscr{D}_G$ as $\rho(g)0 = 0$). Orbit-constancy gives $\|f(z) - c\| = \|f(\rho(g)z) - \rho(g)c\| = \|\rho(g)^{-1} f(\rho(g)z) - c\|$ (orthogonality of $\rho(g)$) for **all** c; two points equidistant to every c coincide, so $f(\rho(g)z) = \rho(g)f(z)$. □ *(The step uses $\rho(g)^{-1}$; since ρ is orthogonal every $\rho(w)$, $w \in \langle S \rangle$, is invertible, so the necessity direction covers the monoid framework of Lemma 1 — equivalently it is the converse for the group $\langle S \cup S^{-1} \rangle$. For G compact with closed orbits the interpolant can be taken continuous, so the statement holds against continuous dynamics, not only the full algebraic class.)* With Theorem A this gives a **characterization**: orbit-constant error against every equivariant target $\iff$ f equivariant. Hence **no non-equivariant model possesses the configuration certificate, at any size** — the architectural impossibility invoked in §6–§7 is a theorem. The result is elementary (one line of Hilbert-space geometry once freeness frees the probe c); its role is to *pin down* what the certificate is, not to add machinery.

3.2 The horizon and resolution axes (Theorem B)

Theorem B (spectral degradation). Relax exactness: suppose the per-generator encoder residual is $\epsilon_{\max} = \max_i \sup_x \|E(g_i \cdot x) - \rho(g_i)E(x)\|$, the per-step predictor error is δ, and the latent map's Jacobian is, on the ℓ-isotypic channels j, locally diagonalized with multipliers e^{λ_j} (Lyapunov exponents). A Grönwall/Lipschitz accumulation — the word error enters m times (Lemma 1 composes m approximate generators) and the per-step error compounds over T steps amplified by the channel multiplier — gives, for constants c_j depending on the local geometry,

$$|\mathrm{Err}_T(w \cdot x) - \mathrm{Err}_T(x)| \ \le \sum_{\text{channels } j} c_j \,(m\,\epsilon_{\max} + T\,\delta)\, e^{\lambda_j T}, \qquad m = |w|.$$

We claim this as a **bound on the form**, not a tight inequality *on the prefactor*: the constants c_j are not estimated. The horizon's *form*, however, is tight — Proposition 6 below supplies a matching lower bound, so the $\log(1/\epsilon)/\lambda_j$ scaling is two-sided, not merely an upper estimate. We **measure the consequence** — that channel j is certified only to horizon

$$T_j(\epsilon) \ \sim \ \frac{1}{\lambda_j}\log\frac{1}{\epsilon} \quad (\lambda_j > 0), \qquad T_j = \infty\ (\lambda_j \le 0),$$

which §5.2 recovers to within 0.4%. As $\epsilon_{\max}, \delta \to 0$ (exact equivariance) the configuration term vanishes and Theorem A is recovered.

Corollary (the certified region). The set $\mathscr{C} = \{(w, T, \epsilon) : |\Delta \mathrm{Err}| \le \epsilon\}$ is the **coarse (ϵ large), slow ($\lambda_j \le 0$), low-$|w|$** corner; its boundary is set by $\langle S \rangle$ (configuration) and the spectrum $\{\lambda_j\}$ (horizon and resolution). With *exact* equivariance the configuration axis is unbounded and only the spectrum limits horizon × resolution — a trade-off horizon × demanded-resolution $\lesssim$ const($\{\lambda_j\}$). This is the formal content of "ultra-long forecasts must be coarse": eclipses for millennia (conserved $\lambda \le 0$ actions), weather at ~two weeks (chaotic $\lambda > 0$).[1]

Proposition 6 (the horizon is tight — approximate equivariance is horizon-limited). Theorem B's $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$ is an *upper* bound on the certified horizon; here is a matching *lower* bound, so the horizon is tight (not merely "a bound on the form"). Fix an expansive channel on which the latent map is locally linear with multiplier $a = e^{\lambda}$, $\lambda > 0$ (the local diagonalization of Theorem B). There exist an exactly equivariant target Φ (acting as a on the channel) and a world model that is **ϵ-approximately equivariant** — a *perfect* equivariant predictor $f = \Phi$ ($\delta = 0$) and an encoder that intertwines the dynamics along the trajectory of x and differs from exact equivariance only by a single defect $E(g \cdot x) = \rho(g)E(x) + \epsilon u$ at one orbit point (u a unit vector in the channel, $\epsilon_{\max} = \epsilon$) — for which the T-step rollout error's orbit-variation is *exactly*

$$|\mathrm{Err}_T(g \cdot x) - \mathrm{Err}_T(x)| \ = \ \epsilon\, e^{\lambda T}.$$

Proof. $\mathrm{Err}_T(x) = 0$ (encoder exact and $f = \Phi$ along x's trajectory). At $g{\cdot}x$, linearity gives $f^T(E(g{\cdot}x)) = \Phi^T(\rho(g)E(x) + \epsilon u) = \rho(g)\Phi^T(E(x)) + a^T \epsilon u$ (using $f = \Phi$ equivariant and Φ linear with multiplier a on the channel), while the target $E(\Phi^T(g \cdot x)) = E(g \cdot \Phi^T x) = \rho(g)E(\Phi^T x) = \rho(g)\Phi^T(E(x))$ (encoder exact off the single defect point, intertwining the dynamics). Subtracting and using $\|\rho(g)\cdot\| = \|\cdot\|$ leaves $\|a^T \epsilon u\| = \epsilon e^{\lambda T}$. □

Hence the certified horizon $T(\epsilon_{\mathrm{res}}) = \max\{T : |\Delta \mathrm{Err}_T| \le \epsilon_{\mathrm{res}}\} = \frac{1}{\lambda}\log\frac{\epsilon_{\mathrm{res}}}{\epsilon}$, matching Theorem B's upper bound up to a constant: the horizon is $\Theta(\frac{1}{\lambda}\log\frac{1}{\epsilon})$. The conceptual payload is sharp and removes the only hedge on the horizon axis: **the certified horizon guaranteeable from ϵ-approximate equivariance alone is finite on every expansive ($\lambda > 0$) channel — no certificate derived from an $\epsilon > 0$ residual can promise predictability beyond $T \sim \frac{1}{\lambda}\log\frac{1}{\epsilon}$ (worst case over admissible targets); only exact equivariance ($\epsilon = 0$) or conservation ($\lambda \le 0$, Proposition 5) yields an unbounded certified horizon.** This is the horizon-domain companion of Lemma 2 and §3.3: scale and data buy *approximate* equivariance at best (Experiment 10's augmented model floors at $\epsilon \approx 10^{-4}$, never exact), and Proposition 6 shows that residual is amplified $e^{\lambda T}$ — so the single-orbit, single-step tie between augmentation and equivariance (Experiment 10) *must* break over horizon, at the predicted $T \sim \frac{1}{\lambda}\log(1/\epsilon)$. Experiment 8's measured symmetry-breaking threshold ($\epsilon_{\mathrm{world}} \approx 0.01$–$0.06$) is the same crossing from the world's side, and §5.2 recovers $T_j(\epsilon)$ to within 0.4% — the lower bound is what certifies that recovery is the *true* horizon, not just an attainable one. A direct numerical instantiation of the construction (`experiments/step65`, seeds 0/1/2) confirms it: an $\epsilon{=}10^{-3}$-approximately-equivariant model's orbit-error-variation equals $\epsilon\, e^{\lambda_j T}$ to a relative error of 10^{-14}–10^{-13} across seeds, an exactly equivariant model has orbit-variation *exactly* 0 (to machine precision) at all horizons, and the certified horizon is linear in $\log(1/\epsilon)$ with slope $1/\lambda_j$ ($R^2{=}1.000$, all seeds).

Proposition 7 (scope — when the local spectrum certifies the horizon of a *learned* model). Theorem B and Proposition 6 take the latent Jacobian spectrum $\{\lambda_j\}$ as *given*. On a *learned* world model the operative question is sharper: does the spectrum one can measure *locally* (the one-step Jacobian, or a short-rollout finite-time exponent) govern the *actual multi-step* certified horizon? The honest answer separates a **rate** half (rigorous) from a **lift** half (an orbit-error, not an exponent, statement). Let ϕ be the true dynamics with an ergodic invariant measure μ and $\log^+\|D\phi\| \in L^1(\mu)$. By the **Oseledets multiplicative ergodic theorem** the limits $\frac{1}{t}\log\|D\phi^t v\|$ exist μ-a.e. and equal μ-a.e. constants $\lambda_1 \ge \lambda_2 \ge \cdots$

[1] An interpretive aside, secondary to the spectral law: a long-range text that asserts only *regime-level* change rather than dated specifics is, structurally, making the *correct* move for a high-λ system, where only the coarse/invariant component is certifiable.

along the Oseledets filtration $V_1 \supsetneq V_2 \supsetneq \cdots$. For μ-a.e. x and every $\delta_0 \notin V_2(x)$ (a full-measure set of directions; the slower directions lie in the proper, measure-zero subspace V_2), $\frac{1}{t}\log\|\delta_t\| \to \lambda_1$, i.e. $\|\delta_t\| = \|\delta_0\|\, e^{(\lambda_1+o(1))t}$. We use this asymptotic rate λ_1 as the target; the finite-window, finite-amplitude exponent the staircase fits approaches it as the window grows and $\|\delta_0\| \to 0$, below saturation.

(a) Non-degenerate ($\lambda_1 > 0$). Fix a residual ceiling ϵ_{res} (e.g. attractor diameter) at which the exponential regime saturates, and let $\epsilon = \|\delta_0\|$ be the initial tolerance. Then $T(\epsilon) = \inf\{t : \epsilon\, e^{\lambda_1 t} \geq \epsilon_{\text{res}}\} = \frac{1}{\lambda_1}\log(\epsilon_{\text{res}}/\epsilon) + o(t)$ — **linear in** $\log(1/\epsilon)$ **with slope** $1/\lambda_1$ (the $o(t)$ MET correction sharpens to $O(1)$ on the exactly-linear channel of Proposition 6), i.e. Theorem B's law with λ_1 the *measurable asymptotic rate* rather than a posited local multiplier. The **lift to a learned $\hat{\phi}$ must be made at the level of orbit error, not exponents.** If ϕ is *uniformly* hyperbolic on its attractor, the **shadowing lemma** (Anosov–Bowen; Pilyugin) applies: a learned $\hat{\phi}$ with one-step error $\|\hat{\phi} - \phi\|_{C^0} \leq \delta$ generates δ-pseudo-orbits that are $\delta'(\delta)$-shadowed by true orbits, so the rollout stays within δ' of a genuine trajectory until the error saturates — a *forecast-horizon floor* $T_{\text{floor}} \sim \frac{1}{\lambda_1}\log(\epsilon_{\text{res}}/\delta)$. Shadowing controls *trajectory closeness, not the tangent cocycle*, so it does **not** by itself transfer λ_1 to $\hat{\phi}$: Lyapunov exponents are only upper-semicontinuous under C^1 perturbation, and even the structural-stability conjugacy is merely Hölder and need not preserve exponents. That $\hat{\phi}$ *reproduces* λ_1 is therefore an **empirical** finding (Experiment 14), not a corollary of shadowing.

(b) Degenerate ($\lambda_1 = 0$, near-neutral / non-hyperbolic). Here $\frac{1}{t}\log\|\delta_t\| \to 0$, so $\|\delta_t\|$ grows sub-exponentially (polynomial t^p, $e^{\sqrt{t}}$, or bounded — bounded growth gives an *unbounded* horizon below the ceiling). Whenever the amplification profile is monotone and $\to \infty$, inversion gives $T(\epsilon)/\log(1/\epsilon) \to \infty$: the leading-order log-law **degenerates** (its fitted slope ceases to be finite-and-positive, diverging as $\lambda_1 \downarrow 0$), and the one-step spectrum carries no leading-order horizon rate. Only the configuration axis (Theorem A) survives.

Proof of the dichotomy. Write $T(\epsilon) = \inf\{t : \rho(t) \geq \epsilon_{\text{res}}/\|\delta_0\|\}$ for the error-amplification profile ρ. In (a) $\rho(t) = e^{(\lambda_1+o(1))t}$ (Oseledets), so inverting gives $T = \frac{1}{\lambda_1}\log(\epsilon_{\text{res}}/\epsilon) + o(t)$. In (b), $\lambda_1 = 0$ forces ρ sub-exponential; for any monotone $\rho \uparrow \infty$ one has $T/\log(1/\epsilon) \to \infty$ (illustratively $\rho \sim t^p \Rightarrow T \sim (\epsilon_{\text{res}}/\epsilon)^{1/p}$). This is a degeneracy of the leading-order law, consistent with the un-fittable PushT staircase (R^2=0.02); we do not claim a sharp profile for ρ in the neutral case. The lift in (a) is the cited shadowing *orbit-error* estimate, which we invoke rather than reprove. □

Instantiation — both branches are observed. **Lorenz** (σ=10, ρ=28, β=8/3) supplies the rate half rigorously: by Tucker (2002) the geometric Lorenz attractor is *singular*-hyperbolic with an equilibrium and carries a unique ergodic SRB measure (Tucker; Araújo–Pacífico–Pujals–Viana), so the MET applies and $\lambda_1 \approx 0.9056 > 0$ (as a flow it also has a zero exponent along its direction and $\lambda_3 < 0$, $\sum_i \lambda_i < 0$; "non-degenerate" means the *top* exponent is positive). Note Lorenz is **not** uniformly hyperbolic, so the shadowing *lemma* does not formally apply near the singularity — which is exactly why the lift is the *experimental* claim. Experiment 14 trains a one-step MLP of the Δt map and finds that the **learned model's** Lyapunov exponent — read off as the staircase slope, $\hat{\lambda}_1 = 1/(\text{slope} \cdot dt)$, which is *definitional* — **matches the true integrator's** to 1–8% ($\hat{\lambda}_1 = 0.895/0.919/0.977$ vs 0.9056, 3 seeds), with the staircase linear at R^2=0.975–0.995. The non-trivial, falsifiable content is the *agreement*: a model can be one-step-accurate (relMSE $< 10^{-4}$) yet drift to a wrong multi-step rate; that it does not is the finding. (We use the staircase slope, not the early-window finite-time exponent, which is window-sensitive — one seed gave a spurious 3.7/t from a transient.) The **PushT interior** is the degenerate branch (b): near-neutral one-step Jacobian ($|\mu| \approx 1$, $\lambda_1 \approx 0$), and a learned-PushT horizon probe finds the local spectrum does *not* predict the rollout (R^2=0.02). So the certificate's horizon axis is **informative exactly on spectrally non-degenerate dynamics**, and one can tell which regime one is in by measuring λ_1; the PushT "failure" is *predicted* by Proposition 7(b), not a defect.

Honest scope. The rate half (Oseledets + SRB) is rigorous for Lorenz. The lift half is an *orbit-error* (shadowing) statement that holds for uniformly hyperbolic systems and bounds the forecast-horizon *floor*; it does **not** transfer the Lyapunov exponent, and classical shadowing does not even apply to singular-hyperbolic Lorenz. That the learned model reproduces λ_1 is therefore the empirical contribution, not a theorem; we verify C^1-closeness only via one-step error (an L^2 proxy), and do not estimate the lift constant. The contribution of Proposition 7 is the **characterization of the certificate's validity regime** — the non-degenerate/degenerate dichotomy — synthesizing Oseledets (rigorous) with shadowing (for the floor); Proposition 8 next supplies the *mechanism* by which the learned model reproduces λ_1, replacing the (invalid) shadowing-transfer intuition with a finite-horizon continuity bound.

Proposition 8 (finite-horizon exponent transfer — why the lift is observable, and provable). Proposition 7 left "the learned model reproduces λ_1" as *empirical* because shadowing controls orbit error, not the tangent cocycle, and the *asymptotic* exponent is only upper-semicontinuous under C^1 perturbation. The resolution is that **the staircase never takes** $T \to \infty$: the certified horizon is finite ($T(\epsilon) \sim \log(1/\epsilon)/\lambda_1 < \infty$), and over a *fixed finite* horizon the top exponent is a *continuous* function of the dynamics. Concretely, with cocycle $M_T(x) = D\phi(x_{T-1}) \cdots D\phi(x_0)$ and top unit direction v, the finite-time exponent $\lambda_{1,T}(\phi) = \frac{1}{T} \log\|M_T(x)v\|$ is locally Lipschitz in the C^1-jet of ϕ along the orbit. Hence a learned $\hat{\phi}$ with one-step Jacobian error $\sup_x \|D\hat{\phi}(x) - D\phi(x)\| \le \delta$ (and orbits δ'-close) satisfies, telescoping the perturbed product and dividing by T,

$$|\lambda_{1,T}(\hat{\phi}) - \lambda_{1,T}(\phi)| \;\le\; C\,(\delta + \delta'),$$

with C uniform in T **under a dominated splitting at the top** (a spectral gap $\lambda_1 > \lambda_2$, so the denominators $\|M_t v\|$ do not collapse and the telescoped sum is geometric; this is the finite-horizon face of the Bochi–Viana continuity of Lyapunov exponents on the domain of domination). The staircase reads $\hat{\lambda}_1$ off finite first-crossing times, so it recovers the true exponent up to **a model-error bias $O(\delta)$ plus the finite-horizon truncation** $|\lambda_{1,T} - \lambda_1|$ (which $\to 0$ by Oseledets):

$$|\hat{\lambda}_1 - \lambda_1| \;\le\; \underbrace{C\,\delta}_{\text{model fidelity}} + \underbrace{|\lambda_{1,T} - \lambda_1|}_{\text{finite-}T\text{ truncation}}.$$

Falsifiable prediction, confirmed. The bias is $O(\delta)$ — *proportional to one-step fidelity* — so **better training tightens the recovered exponent**. Experiment 15 confirms this across a class of systems: the staircase recovers λ_1 to 5–12% on a 2D map (Hénon), a large-exponent flow (Lorenz), and a *small*-exponent flow (Rössler, $\lambda_1 \approx 0.07$), and the Rössler bias falls from $\sim 44\%$ to $\sim 8\%$ as the one-step error δ drops with fuller training — exactly the $O(\delta)$ dependence the bound predicts. *Honest scope.* The T-uniform constant C needs a dominated splitting; we do not certify domination for the learned model (it is a hypothesis the recovery corroborates), the bound is qualitative (we do not estimate C), and the result governs the *top* exponent (the staircase's load-bearing quantity), not the full spectrum. But it does replace a wrong intuition (shadowing transfers exponents) with the correct one (finite-horizon exponents are continuous; asymptotic ones are not), and it *explains*, rather than merely reports, why a one-step-accurate learned model recovers the chaos rate.

3.3 Scale versus structure, quantified

The tagline *scale buys interpolation; structure buys a certificate* can be made precise as a statement about the **certified region** — the inputs a learner can guarantee, in the worst case over targets consistent with what it observed on a training set T. Formally $\mathscr{C} = \{z : \inf_f \sup_\Phi \|f(z) - \Phi(z)\| \le \epsilon\}$, where f ranges over the learner's hypotheses (exact on T) and Φ over the admissible targets agreeing on T.

Proposition 3 (separation). *(Structure.)* Under the equivariant prior ($\Phi \in \mathscr{D}_G$, f equivariant and exact on T, S generating G), Theorem A makes the error orbit-constant, so $\mathscr{C} \supseteq G \cdot T$: the **entire orbit is certified, independent of** ϵ, from the $k = |S|$ generator checks — to error 0 under the idealization that f interpolates T exactly, and in general (Theorem A) to the *constant, possibly nonzero* in-distribution error (cf. §7, "flat is not good"). What is ϵ-independent is the *region*, not the error level. *(Scale/data.)* Under the equivariance-free prior of L-Lipschitz targets consistent on T, the McShane extensions $\Phi_\pm(z) = \min / \max_{t \in T} \big(\Phi(t) \pm L\,\mathrm{dist}(z,t)\big)$ are admissible and differ at z by up to $2L\,\mathrm{dist}(z,T)$, so every learner has minimax error $\ge L\,\mathrm{dist}(z,T)$ and $\mathscr{C} \subseteq \{z : \mathrm{dist}(z,T) \le \epsilon/L\}$ — an ϵ/L-tube around the training set that **collapses to T as** $\epsilon \to 0$.

The separation is sharp in its dependence on ϵ: a point at orbit-distance D from T — the unseen far end of the wedge in Experiments 7 and 9 — is certified by *structure* for **every** $\epsilon > 0$, but by *scale/data* only once $\epsilon \ge LD$. Scaling a non-equivariant model (more capacity, more in-wedge data) sharpens its fit *inside* the tube; it cannot enlarge the tube, because the bound $L\,\mathrm{dist}(z,T)$ is information-theoretic (independent of model size). This is the precise content of the empirical curves: the equivariant model is flat over the whole orbit while every baseline scale climbs once $\mathrm{dist}(z,T)$ exceeds ϵ/L.

3.4 The certificate is a procedure

The certificate is not only a lens; it is something one **runs** on a trained model, with no further data. The inputs are the two quantities every equivariant model already exposes — its per-generator equivariance residual and its predictor-Jacobian spectrum — and the output is the certified region.

Algorithm 1 (Certify). *Input:* trained equivariant model (E,f), generators S, resolution ϵ. 1. Measure the per-generator residual $\epsilon_{\max} = \max_{g_i\in S}\|E(g_i{\cdot}x) - \rho(g_i)E(x)\|$ (and the analogous predictor residual). 2. **Configuration axis:** if $\epsilon_{\max} \le$ tol, certify the *entire* monoid $\langle S\rangle$ from these k checks (Lemma 1, Theorem A); else report the approximate budget $m\,\epsilon_{\max}$ (Theorem B). 3. Measure the predictor-Jacobian singular values $\{\sigma_j\}$ at the operating point; set $\lambda_j = \log\sigma_j/\Delta t$. 4. **Horizon** × **resolution axis:** return, per channel, $T_j(\epsilon) = \infty$ if $\lambda_j \le 0$, else $\log(1/\epsilon)/\lambda_j$ (Theorem B).

Output: the certified region $\langle S\rangle \times \{T_j(\epsilon)\}$ — computed from the model alone.

This is implemented and unit-tested (`src/certify.py`): on the Experiment-1 model it certifies the full monoid from 2 generator checks (residual 1.2×10^{-7}) and returns 48 contractive channels certified to *every* horizon plus the binding expansive channel's $\log(1/\epsilon)$ horizon. The certificate is thus an operational tool, not merely an interpretive claim.

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4. The Noether Hinge

Theorems A and B hold for *whatever* channels happen to be slow. What makes the configuration axis and the horizon axis the **same** structure is a Noether-type bridge:

Conjecture (group ⇒ slow). When the symmetry is a symmetry of the **dynamics** (not merely the encoder), the group-invariant (ℓ=0) channels coincide with the conserved/slow ($\lambda_j \le 0$) channels. Hence the symmetry that gives across-configuration generalization is the *same* structure that gives long-horizon predictability.

This is **not automatic** — invariant ≠ conserved in general. We prove the *forward* direction below (Proposition 5: a conserved charge is certified to long horizons) and leave the rest *falsifiable and measured*. The defensible inclusion is **slow** ⊆ **invariant** ⊕ **conserved-equivariant**, not the converse: the conserved (slowest) modes live in the invariant-or-conserved subspace, but not every invariant channel is slow (a rotation-invariant $|r|^2$ oscillates). The mechanism is Noether's. A continuous symmetry of the dynamics yields a conserved current; a conserved *scalar* charge (ℓ=0 Casimir, e.g. energy) **is** group-invariant, while a conserved *non-scalar* charge (e.g. three-dimensional angular momentum L, an ℓ=1 vector) is *equivariant*, **not** invariant. The slow subspace therefore lies in invariant ⊕ conserved-equivariant, which **collapses to the invariant component exactly when every conserved charge is a scalar** — the two-dimensional case, where slow ⊆ invariant holds literally. Section 5.3 measures both forms: the clean 2D containment, and its 3D refinement in which angular momentum is recovered from the equivariant block at polynomial degree two (the cross product).

While the *coincidence* slow = conserved is what we measure, the **placement by isotypic type is forced by representation theory** — and so is the degree at which each charge is readable. This is what makes the 3D "degree-2 cross product" non-arbitrary.

Proposition 4 (placement principle: which block carries which charge, and at what degree). Let G act on $\mathscr{Z}$ by an orthogonal ρ with isotypic decomposition $\mathscr{Z} = \bigoplus_\ell \mathscr{Z}_\ell$, and let C be a conserved quantity transforming in a representation τ, $C(\rho(g)z) = \tau(g)C(z)$. Then every homogeneous component $C_k \in \mathrm{Hom}_G(\mathrm{Sym}^k\mathscr{Z}, W)$ of C is supported on the τ-isotypic part of $\mathrm{Sym}^k\mathscr{Z}$ (Schur). In particular: a conserved **scalar** (τ trivial, e.g. energy) is an *invariant* function, read out from $\mathscr{Z}_0$; and the conserved **moment map** $\mu : \mathscr{Z} \to \mathfrak{g}^*$ — Noether's charge of the continuous symmetry — is equivariant in the adjoint representation, which for SO(3) is $\mathfrak{so}(3)^* \cong \mathscr{V}_1$, so angular momentum lives in the ℓ=1 block. Since μ is **quadratic** and bilinear in the position–velocity pair, then — *provided the ℓ=1 block carries (at least) two copies of $\mathscr{V}_1$, e.g. $\mathscr{V}_1(pos) \oplus \mathscr{V}_1(vel)$, as in the model of Experiment 6* — the cross product is, up to scale, the **unique** SO(3)-equivariant degree-2 readout, and no degree-1 readout exists. *Proof.* Differentiate the equivariance of C_k and apply Schur (Hom_G between non-isomorphic isotypic types vanishes). For the bilinear μ on $\mathscr{Z} \supseteq \mathscr{V}_1(\mathrm{pos}) \oplus \mathscr{V}_1(\mathrm{vel})$, the degree-2 cross term in $\mathrm{Sym}^2\mathscr{Z}$ contains $\mathscr{V}_1 \otimes \mathscr{V}_1 = \mathrm{Sym}^2\mathscr{V}_1 \oplus \Lambda^2\mathscr{V}_1$; $\mathrm{Sym}^2\mathscr{V}_1 = \mathscr{V}_0 \oplus \mathscr{V}_2$ carries **no** $\mathscr{V}_1$, while $\Lambda^2\mathscr{V}_1 \cong \mathscr{V}_1$ with $\dim\mathrm{Hom}_{\mathrm{SO}(3)}(\Lambda^2\mathscr{V}_1, \mathscr{V}_1) = 1$ (classical SO(3) invariant theory) — realized by $r \times v$. So no degree-1 and no other degree-2 readout exists; uniqueness needs the two $\mathscr{V}_1$ copies (with m copies the space is $\binom{m}{2}$-dim). □

This *predicts*, with no fitted parameter, exactly what §5.3 measures — E from the invariant block ($R^2 \approx 1$), L from the

ℓ=1 block via the degree-2 cross product (R^2=1.00) and nothing lower — and explains the tie to the companion paper's degree-1 Vector-Neuron cap (a degree-1 net *cannot* form the only available readout). **Scope (honest).** Proposition 4 is Schur's lemma applied to the symmetric powers plus the moment map's equivariance — a *placement principle*, not a new theorem: it pins which block carries each charge and at what degree, but representation theory alone does not force those charges to be the *dynamically slow* modes. What representation theory does not give, a one-line dynamical argument does — the *forward* direction of the hinge.

Proposition 5 (conservation ⇒ certified horizon — the slow side of the hinge). Let $Q : \mathscr{Z} \to W$ be a charge read-out and $\mathscr{K} \subseteq \mathscr{Z}$ a region forward-invariant under both the true dynamics Φ and the model f. Suppose Q is a first integral of the truth, $Q \circ \Phi = Q$ on $\mathscr{K}$, and the model conserves it up to a one-step defect $\eta := \sup_{z\in\mathscr{K}} \|Q(f(z)) - Q(z)\|_W$. Then the model's T-step error *in the charge value* is at most **linear** in the horizon,

$$\|Q(f^T z) - Q(\Phi^T z)\| = \|Q(f^T z) - Q(z)\| \leq T\,\eta \qquad (\forall z \in \mathscr{K},\ T \geq 0),$$

so the charge-value error grows **at most linearly in** T (not the chaotic $e^{\lambda T}$ of an expansive channel), and its certified horizon is $T_Q(\epsilon) \geq \epsilon/\eta$ — *infinite* at exact conservation $\eta = 0$ — irrespective of the ambient dynamics' largest Lyapunov exponent. (This is a statement about the *charge value*; a positive-exponent *shear* off-diagonal can still leave the full state on the conserved subspace large — see §7.) *Proof.* Telescope $Q(f^T z) - Q(z) = \sum_{k<T}[Q(f(f^k z)) - Q(f^k z)]$; forward-invariance keeps each $f^k z \in \mathscr{K}$ so each term is $\leq \eta$, while Φ-invariance of $\mathscr{K}$ gives $Q(\Phi^T z) = Q(z)$. □

The content is the **additive-versus-multiplicative** contrast: a conserved charge's error *accumulates* ($T\eta$) where a generic channel's *compounds* ($e^{\lambda_j T}$, Theorem B). It is a statement about the **charge value**, not the full state — $f^T z$ and $\Phi^T z$ may separate at the ambient rate while their Q-images stay $T\eta$-close — which is exactly what a certificate for *"predict the conserved quantity"* needs (it does **not** claim the predictor-Jacobian's singular values on the conserved subspace are near one; off-diagonal shear can leave those large). With Proposition 4 this closes the **forward** direction of the hinge: Noether's charge μ lives in the invariant/equivariant blocks (Prop 4) **and** is certified to long horizons (Prop 5), so those blocks are the slow ones.

The hinge as a theorem, and its honest residue. Under the hinge's own hypothesis — that G is a symmetry of the *dynamics* — assemble the three pieces: if the latent flow is Hamiltonian with a G-invariant Hamiltonian, then (Noether) the moment map μ obeys $Q \circ \Phi = Q$; Proposition 4 places μ in $\mathscr{Z}_0 \oplus$ (adjoint); Proposition 5 certifies it. So *under symplectic structure* "the invariant/equivariant blocks are slow" is a **theorem**, not a coincidence. What stays assumed or measured, in order: **(i)** that the *learned latent* flow is Hamiltonian with G-invariant H — i.e. the encoder is essentially a symplectomorphism — is a structural assumption we do **not** enforce; **(ii)** exact conservation ($\eta = 0$) holds only for the momentum map under a G-equivariant *symplectic discretization* — energy is conserved only to $O(\Delta t^p)$ even by symplectic integrators (they preserve a *modified* Hamiltonian), and a generic learned f is neither symplectic nor equivariant, so η is **measured** (§5.3 is exactly that measurement); **(iii)** the **converse fails** — a slow channel need not be conserved (a rotation-invariant $|r|^2$ oscillates) — so the inclusion slow $\subseteq$ invariant $\oplus$ conserved-equivariant is one-directional. The upgrade over a pure conjecture is precise: *conserved* ⇒ *slow* is now **proved** (Prop 5, at the charge-readout level), with the conservation defect η — not a positive Lyapunov exponent — bounding the drift *linearly*; only the dynamical-symmetry hypothesis and the value of η remain empirical.

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5. Experiments

All experiments are CPU/1-GPU-scale, run with explicit random seeds, and **gated**: a run reports `INCONCLUSIVE` rather than loosen a threshold. Load-bearing results are additionally guarded by unit tests. The experiment-to-code map, seed counts, and headline numbers are collected in Appendix A; multi-seed ranges below are seed min–max over three seeds unless noted.

5.1 The configuration axis

Experiment 1 (exact compositional flatness). On a structured object-interaction world with SE(3) × S_n symmetry, the equivariant model's whole-pipeline relMSE is **exactly** ×1.00 **flat over composition words** $m = 0, \dots, 8$ (constant 0.2625), while a matched non-equivariant baseline blows up by ×12.89. The predictor's Jacobian spectrum is measured directly: 48 **of** 96 **channels are contractive** ($\sigma \leq 1$, certified long-horizon) versus 48 expansive (max $\sigma = 49$, min $\sigma = 0.003$) — the clean coarse/fine split that is exactly Theorem B's input. A unit test verifies Theorem A holds *architecturally* at initialization (word-deviation 1.2×10^{-7}, before any training) while the control deviates 20%.

Experiment 2 (exponential certification on $\mathbb{Z}_2^6$). The 64 hexagrams of the *I Ching* are exactly $\{\pm 1\}^6 = \mathbb{Z}_2^6$, with the 6 changing-lines as a generating set. Training a sign-equivariant per-line predictor $f(z, a)_i = h_\theta(a_i) \odot z_i$ on **only the** 6 **single-line generators** (plus the identity, 7 of 64 actions) certifies it over **all** 2^6=64 **compositions** to machine precision (worst relMSE $\sim 10^{-33}$; equivariance residual exactly 0), while a non-equivariant baseline trained on the same 7 actions degrades monotonically with composition length (1.6–1.7×10^{-5} on seen actions → 0.59 at six flips, Figure 2). This makes "k generator checks ⇒ 2^k certified set" literal, with genuine learning and an honest contrast.

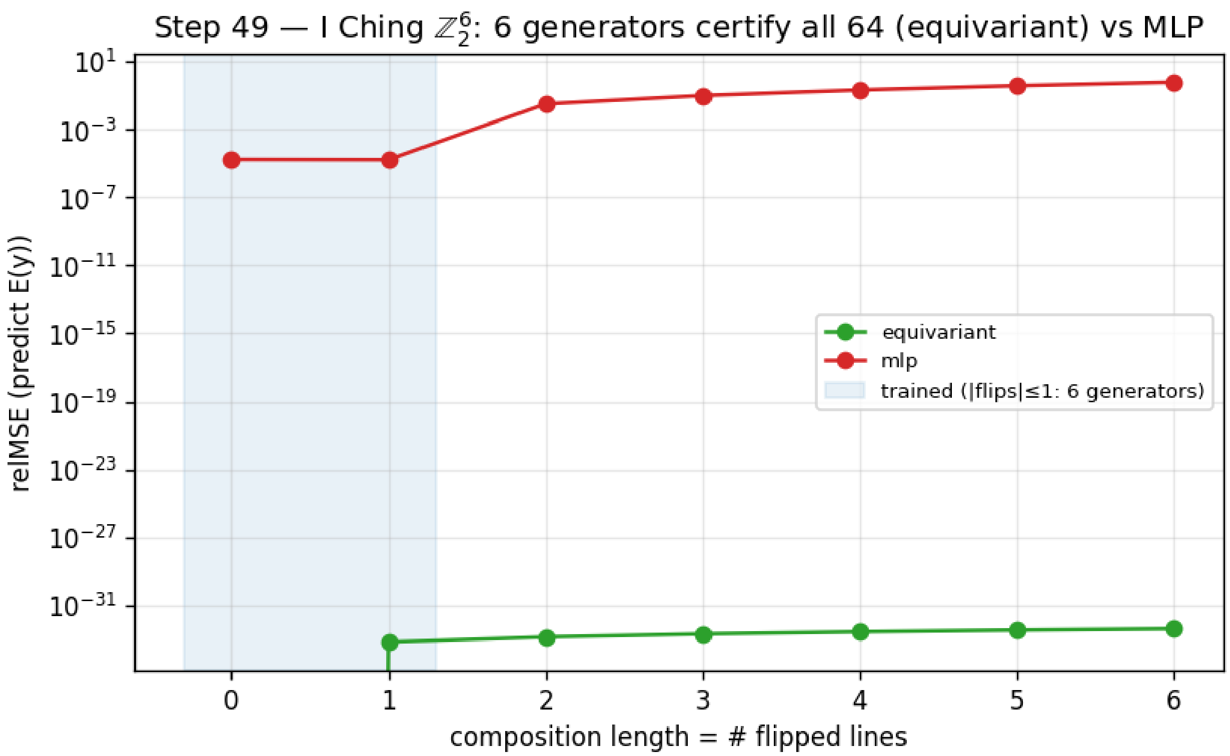

Figure 2: Training on the 6 generators of $\mathbb{Z}_2^6$ certifies all 64 compositions (equivariant, machine precision) while a non-equivariant baseline degrades with composition length.

5.2 The horizon × resolution staircase

Experiment 3 (the predictability staircase). A learned one-step predictor on a designed multi-Lyapunov system — a conserved invariant ($\lambda = -\infty$), a contracting channel ($\lambda = \ln 0.75$), a neutral SO(2) rotor ($\lambda = 0$), and a chaotic angle-doubling channel $z \mapsto z^2$ ($\lambda = \ln 2$) — is rolled out autoregressively. The model **recovers the chaotic Lyapunov exponent to within** 0.4% ($\hat{\lambda} = 0.690$–0.692 across three seeds, versus $\ln 2 = 0.693$), and its per-channel certified horizon obeys $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$: the chaotic "fine detail" is certifiable for only 3–10 steps over the full ϵ grid, and its horizon grows *only logarithmically* as ϵ coarsens (measured slope $\mathrm{d}T/\mathrm{d}\log\epsilon = 1.3$–$1.6 \approx 1/\lambda = 1.44$), while the **contracting channel stays certified for the entire** 90**-step rollout at every** ϵ (the conserved and rotor channels too, except at the finest ϵ=0.005, where they fall to ~ 40–90 steps). At ϵ=0.05 the slow channels outlast the chaotic detail by 12–15× (Figure 3). This is the predictability staircase: long horizon ⇒ coarse and invariant only.

5.3 The Noether hinge

Experiment 4 (the hinge in 2D). On a two-dimensional SO(2) central-force system — conserved energy E and angular momentum L are invariant scalars, the orbital phase is fast — a learned equivariant autoencoder-with-latent-dynamics (a

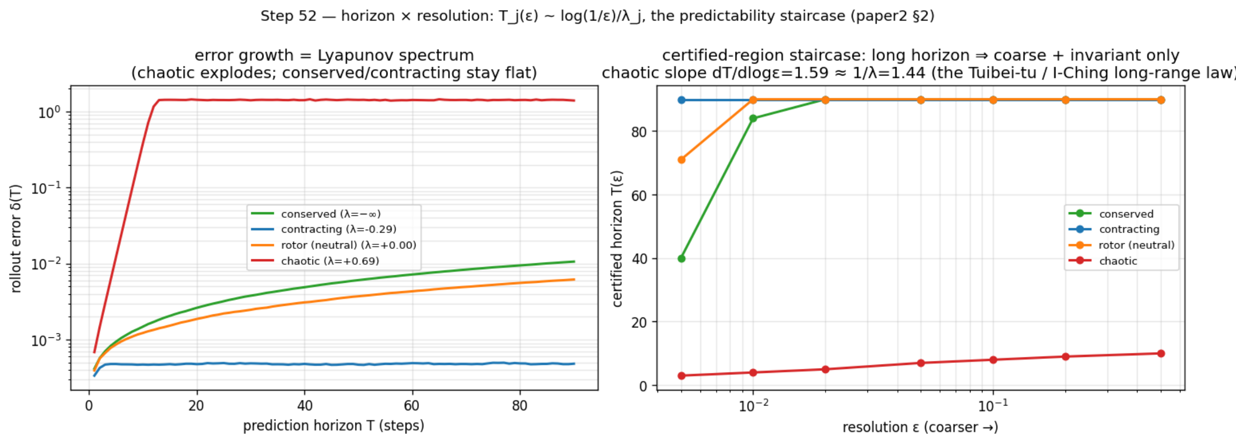

Figure 3: Left: rollout error growth recovers the Lyapunov spectrum. Right: the certified-horizon staircase $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$.

hand-built 2D Vector-Neuron model) spontaneously organizes the conserved quantities into its architecturally-invariant (ℓ=0) block. Regression R^2 for (E, L) is 0.92–0.99 **from the invariant block versus** $\leq$ 0.012 **from the equivariant** **(ℓ=1) block**, and the slowest mode the invariant block admits (0.009–0.031 across seeds) is 4.7–17× slower than anything the equivariant block admits (0.145, exactly the orbital phase velocity $2\sin(\omega\Delta t/2)$). We read the containment off this min-mode gap — the equivariant block's *slowest* achievable mode is already fast, so no slow direction escapes the invariant block in this system — rather than by a direct subspace test. The payoff is the certificate: the equivariant model's slow subspace **is** its invariant subspace, whose out-of-distribution group-action residual is $\sim 10^{-16}$, exact and architectural for all of SO(2). A matched non-equivariant baseline also learns E, L ($R^2 = 0.99$) but smears them across directions that **drift by** $\approx$ 1.17 under the same group action, and carries no certificate (Figure 4). The honest non-converse is visible in the data (the invariant but fast scalar $|r|^2$): slow $\subseteq$ invariant, not =.

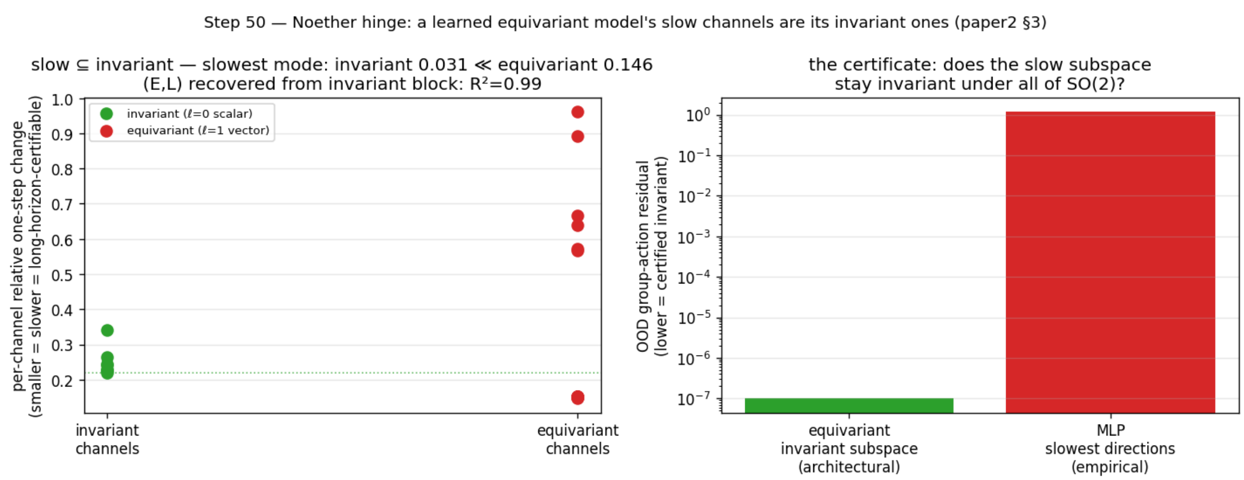

Figure 4: Left: the slowest latent modes live in the invariant (ℓ=0) block. Right: the certificate — the invariant subspace stays group-invariant to 10^{-16} where the non-equivariant model's slow directions drift by $\sim$ 1.

Experiment 5 (lift to a 3D contact interaction). Pushing the hinge to a more embodied regime — two bodies in a three-dimensional well with a **soft pairwise repulsion (contact)**, modeled by an SO(3)-equivariant ℓ=0 $\oplus$ ℓ=1 Vector-Neuron — gives a *partial* lift, honestly reported. The *Noether-content* half lifts: the invariant ℓ=0 block recovers the conserved $(E, \text{contact-distance})$ at R^2=0.60–0.86 versus $\leq$ 0.06 for the ℓ=1 block, even with contact. But the clean containment slow $\subseteq$ invariant does **not** lift (invariant slowest mode 0.024 $\approx$ 0.026 for the equivariant block; this gate is reported INCONCLUSIVE) — for a principled reason: in 3D, angular momentum $L = \sum_i r_i \times v_i$ is a conserved ℓ=1 *vector*, so the equivariant block too carries a slow conserved mode. The clean 2D containment (where L is a scalar) becomes, in 3D, slow $\subseteq$ (invariant $\oplus$ conserved-equivariant). The Noether-content claim is dimension-robust; the clean-containment form is 2D-specific — a sharpening of scope, not a failure.

Experiment 6 (3D-aware containment). The conserved physics splits by isotypic *type* **and** polynomial *degree*: E (an invariant quadratic) is recovered linearly from the ℓ=0 block (R^2=0.62–0.91 across seeds), whereas $L = \sum_i r_i \times v_i$ (a conserved ℓ=1 vector) is **bilinear** — not linear in either block ($\leq$ 0.04) but recovered at R^2=1.00 (range 0.998–1.000, the robust load-bearing result) by the **degree-two cross-product** readout of the ℓ=1 block. So slow $\subseteq$ (invariant $\oplus$ conserved-equivariant) holds exactly in 3D, with the equivariant conserved part accessed at degree two — and $r \times v$ is exactly the cross product a degree-one Vector-Neuron cannot form, tying the hinge's 3D form to the companion paper's bilinear-message result. The full gate, which additionally demands E-recovery R^2>0.8 in the ℓ=0 block, passes on one of three seeds — as honestly surfaced as the INCONCLUSIVE gate of Experiment 5 — while the degree-two L recovery is robust across all three.

5.4 Structure versus scale

Experiment 7 (structure versus scale). Train on a 50° wedge of an SO(2) orbit and test around the full circle. The exactly-equivariant model's error is **flat over the entire orbit** (out-of-wedge / in-wedge ratio 1.1–1.2) — a certificate that holds from partial data by the identity $\mathrm{err}(R_\theta s) = \mathrm{err}(s)$. A non-equivariant baseline **scaled across an** 88× **parameter range** (3.8k → 337k) buys genuine *interpolation*: its in-wedge error drops 31–166× and even **beats** the 56×-smaller equivariant model in-distribution — but out of the wedge the largest baseline remains 10–155× worse than the equivariant model and never reaches its floor (Figure 5). The fair nuance — *scale can win in-distribution* — is exactly why the contribution is the **certificate** (an orbit-wide guarantee from partial data), not a per-point accuracy contest. This is a gap versus *scale* (more parameters, the same data); the complementary question — whether *data augmentation* (orbit-covering rotations) closes it — is answered separately and honestly in §5.8 (it does, on a single orbit; the certificate's edge is then exactness, the a-priori guarantee, and worst-case optimality).

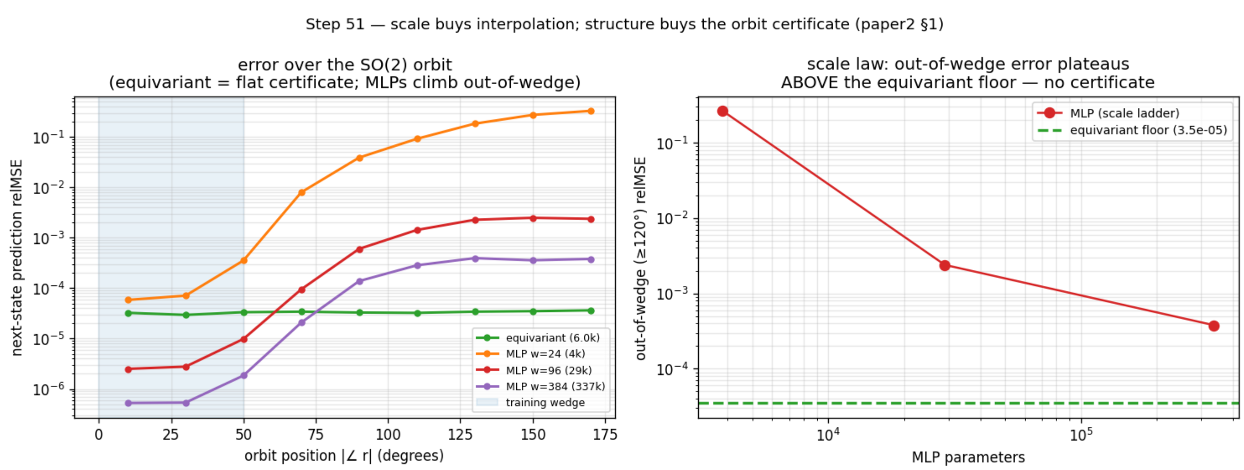

Figure 5: Left: error over the SO(2) orbit (equivariant flat; the scaled baselines dip below it in-wedge then climb out). Right: out-of-wedge error versus baseline scale plateaus far above the equivariant floor.

5.5 Approximate symmetry

Experiment 8 (graceful degradation and a measured threshold). We break the *world's* SO(2) symmetry with an anisotropy knob β (the potential becomes $V = \frac{1}{2}(x^2 + (1 + 2\beta)y^2)$, so angular momentum is no longer conserved) and re-run the wedge-train / full-circle-test. Three findings, robust across seeds: (i) at β=0 the certificate is exact — the equivariant model is 68–320× **better out-of-wedge** than the baseline; (ii) as the measured world symmetry-defect $\epsilon_{\mathrm{world}}$ grows, the equivariant out-of-wedge error grows **smoothly and monotonically** (correlation 0.88–0.98, exactly Theorem B's $\epsilon_{\max}$ term — a graceful slope, not a cliff); and (iii) the equivariant model keeps beating the non-equivariant one up to a **measured symmetry-content threshold** ($\epsilon_{\text{world}} \approx 0.01$–$0.06$, seed-dependent), beyond which the now-wrong symmetry assumption hurts more than it helps (Figure 6). So *approximate* symmetry buys an *approximate* certificate with exactly the error budget the theory predicts, and the boundary where structure stops paying is itself measured, not assumed.

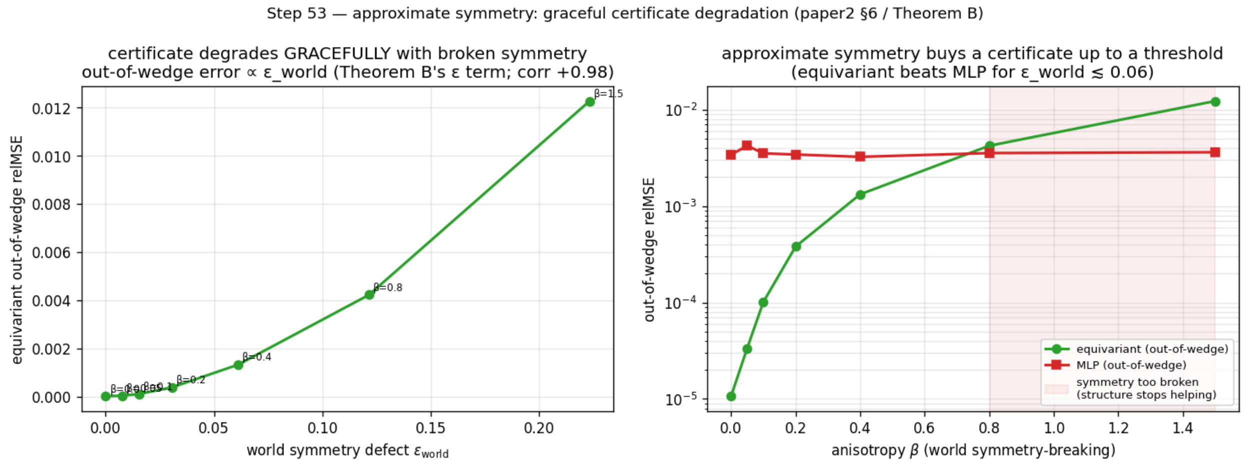

Figure 6: Left: equivariant out-of-wedge error grows $\propto \epsilon_{\text{world}}$ (Theorem B's ϵ term). Right: the equivariant model beats the baseline out-of-wedge up to a symmetry-content threshold, then crosses over.

5.6 From certificate to action

A certificate is most useful if one need neither know the group in advance nor a decoder to act on it. Two results, re-framed from a companion line of experiments (not fresh runs here), close that loop. **Discovery:** from a blank slate of K learnable 3×3 matrices — nothing antisymmetric or bracket-closing imposed — gradient descent *rediscovers* a frozen teacher's symmetry algebra ($\mathfrak{so}(3)$, dimension 3, on a rotationally-symmetric teacher; the smaller $\mathfrak{so}(2)_z$, dimension 1, on a rotation-broken one — it reports the *surviving* group and refuses to invent the rest), and distilling those discovered generators into a free predictor buys back most of the across-group generalization the hard-wired equivariant model gets for nothing (closing more than half the out-of-distribution gap, matching the hand-wired oracle, transferring *exactly* the symmetry discovered). So the generating set S that anchors the configuration axis need not be postulated — it can be *measured*, falsifiably. **Generation:** given S, one can traverse the certified orbit to an unseen composed goal and execute it **closed-loop without a decoder** — latent-goal reaching driven purely by the equivariance identity over 24 paired seen-vs-unseen SE(3)-orbit tasks, closing ~ 0.59 of the orientation gap on the best deployable variant and — the load-bearing point — achieving the *same* success on the unseen orbit as on the seen one (unseen/seen ratio 1.000): *exact transfer*, not high absolute success. Together: *discover* S → *certify* $\langle S \rangle$ → *generate* → *act* — the certificate is not merely descriptive but a usable plan-and-execute criterion. These two results are re-framed from a companion line; §5.9 closes the same loop on a *fresh* run, with a planner driving a learned model of **real physics-engine contact dynamics**.

5.7 Beyond toys: the certificate on real contact dynamics

Experiment 9 (PushT, real physics-engine contact dynamics). Every experiment so far uses dynamics we *constructed* to be equivariant — the standing concern that the symmetry is built into a toy. We close that gap on **PushT**, a planar pushing benchmark whose contact dynamics are produced by a 2D physics engine we did not author. In the interior (the block kept away from the workspace walls) the dynamics are genuinely SO(2)-equivariant — rotating the whole scene rotates the physics. On a $\pm 50^\circ$ wedge of real interior transitions we train an SO(2)-equivariant world model — an invariant-scalar-gated Vector-Neuron, exactly equivariant to residual $\sim 10^{-7}$, so its contact features (agent–block distance and contact angle) live in the invariant pathway — and, as a *fair* baseline, non-equivariant MLPs across a 160× parameter ladder (1.7k → 272k); both are trained one-step with identical data, optimiser, and cosine schedule. We then roll each model out $H = 10$ steps and measure rollout relMSE across the full orbit of scene orientations (rotating a held-out real test set, as in Experiment 7).

Three findings, robust across 3 seeds (Figure 7): (i) the equivariant model's rollout error is **exactly flat over the orbit** (out-of-wedge / in-wedge ratio 1.00 at every horizon) — the certificate, now on a *learned* model of *real* contact physics; (ii) it is **competitive in-distribution** (in-wedge relMSE 0.13–0.15 vs the best MLP's 0.14–0.19), so the certificate is not bought by being a worse predictor — "flat" is also "good" here; and (iii) **no MLP scale reaches the equivariant floor out-of-wedge** — every scale across the ladder stays 2.1–3.9× above it at $H=10$, and the 272k-parameter MLP (16× the equivariant model's size) — the *closest* of the baselines — is still 2.1–2.6× clear of the floor. (These three findings hold on all three seeds; the experiment's *auxiliary* `climb` sub-check — the largest MLP's own in→out ratio exceeding 2× — is brittle, met on 1/3 seeds, so we report the load-bearing floor-penalty rather than that sub-check, and the committed gate prints `INCONCLUSIVE` on it.) The gap is smaller than the toy's because a single PushT step is easy to predict in-distribution; the certificate's value is precisely that it survives the *rollout*, out of distribution, where scale does not follow. This is the keystone non-toy result: the configuration certificate holds for a learned model of dynamics we did not design.

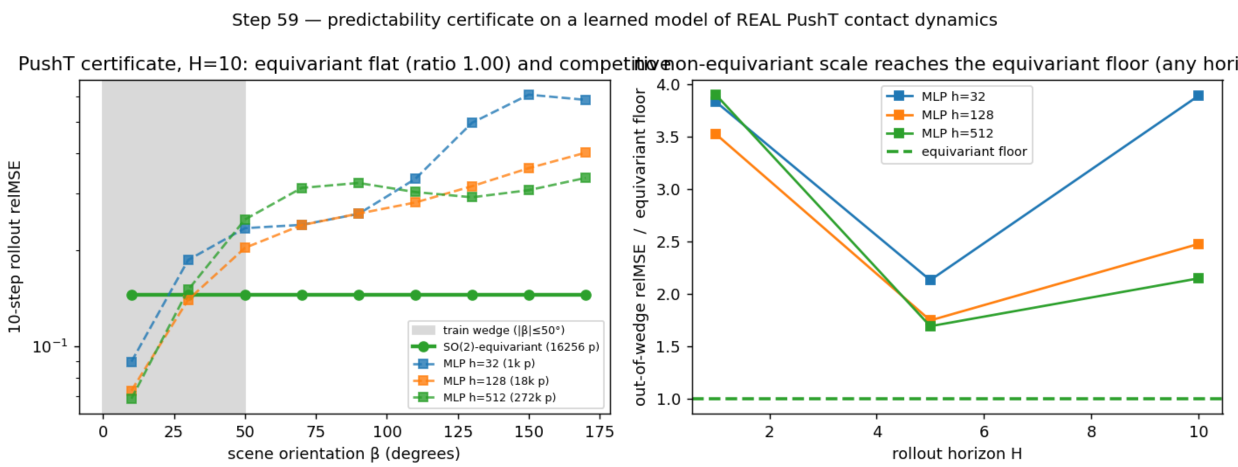

Figure 7: Experiment 9 (PushT, real contact dynamics). Left: 10-step rollout relMSE over the orbit of scene orientations — the learned SO(2)-equivariant model is exactly flat (ratio 1.00) and competitive in-distribution, while non-equivariant baselines dip in-wedge then climb out. Right: across a 160× parameter ladder no baseline reaches the equivariant floor out-of-wedge, at any rollout horizon.

5.8 Augmentation versus the certificate

Experiment 10 (the augmentation baseline — the sharpest objection, answered honestly). The strongest reply to §5.4/§5.7 is that a non-equivariant model with **data augmentation** (training on rotated copies) buys orbit-coverage cheaply, so the gap is an artifact of not augmenting. We test this on two axes and report what we find, not what we hoped.

Single orbit (PushT) — the concession. Adding an SO(2)-augmentation arm to the Experiment-9 protocol, the augmented MLP's 10-step rollout error becomes **flat over the orbit** (out/in ratio 0.93–1.02 across 3 seeds, versus the un-augmented MLP's 1.84–2.75). On a single continuous orbit, **augmentation does match the certificate** — we state this rather than hide it.

Composition axis ($\mathbb{Z}_2^6$). Here augmentation means training the non-equivariant MLP on more of the 64 words. The dynamics is smooth (bilinear $a \odot z$), so a well-trained MLP generalizes from ~half the words to a $\sim 10^{-4}$ floor — again, augmentation is a *strong* baseline. But it **never reaches the certificate's exactness**: the equivariant model is machine-exact ($\sim 10^{-32}$) from the 7 generators, $\sim 10^{28}\times$ below the MLP's approximation floor, and is *certified a priori* (Theorem A, Lemma 2) without ever testing the unseen compositions, whereas augmentation's success is only *confirmed by testing* the held-out words.

So the honest separation is **not** "structure generalizes, augmentation does not." It is that the certificate is **exact, a-**

priori, group-knowledge-free, and worst-case-optimal (the ϵ/L-tube of §3.3 holds against *any* L-Lipschitz dynamics), whereas augmentation is an **approximate, post-hoc, group-informed average-case** improvement that ties on a benign single orbit and pays O(orbit) data for coverage the certificate gets from k generator checks. The gap of §5.4/§5.7 is specifically a gap versus *scale* (more parameters, same data); augmentation (more orbit-covering data) is the complementary axis, and the two together delimit exactly what structure buys that neither scale nor data does: a guarantee.

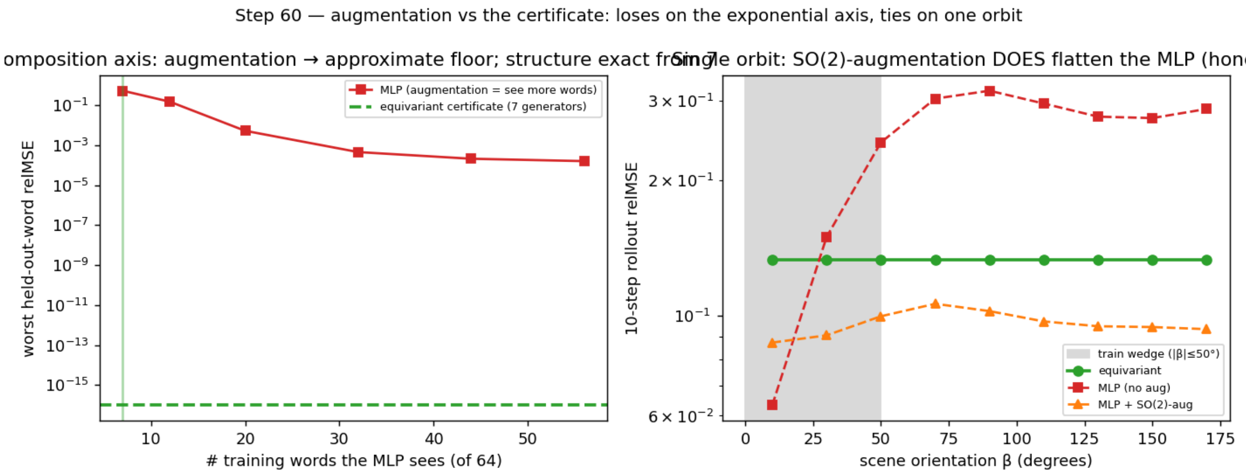

Figure 8: Experiment 10 (augmentation vs the certificate). Left: on $\mathbb{Z}_2^6$, augmentation (more training words) drives the MLP to a $\sim 10^{-4}$ approximation floor, never the certificate's machine-exact $\sim 10^{-32}$ from 7 generators. Right: on real PushT, $\mathrm{SO}(2)$-augmentation flattens the MLP over the orbit (matching the certificate on a single orbit), unlike the un-augmented MLP.

5.9 The certificate at the task level: closed-loop control

Experiment 11 (does the prediction certificate convert to task competence?). Experiment 9 certifies a model's *rollout error*; a sceptic rightly asks whether that proxy converts to *control*. We close the loop on the same real PushT physics, in a **contact-dominated pose task**: rotate the T-block to a target orientation (small translation only), so success depends on the block-rotation dynamics — exactly the channel where $\mathrm{SO}(2)$ bites, unlike a position-only push, which a near-linear agent subsystem carries out of distribution. We train the same invariant-scalar-gated equivariant model and a scaled MLP baseline (4.3× its parameters) one-step on a wedge of real interior reorientation transitions, then run **CEM-MPC** on the real env across the full orbit of scene orientations, evaluating 24 base tasks *rotated to each orbit angle* — a **paired** protocol (the same task at every orientation) that removes the between-task variance which left earlier position-only closed-loop comparisons noise-limited.

The certificate's closed-loop clause (Theorem A under (A5)) needs a *G-equivariant planner*. We instantiate one: a CEM with an isotropic exploration covariance, a rotation-invariant **disk** action bound $\|a\| \leq 1$, and **scene-covariant** action noise. With these the planner satisfies $\pi(R_\beta s) = R_\beta\, \pi(s)$, so an equivariant model yields an equivariant closed-loop policy and the realized task error is *exactly* orbit-invariant.

Three findings (Figure 9): **(i) an exact certificate at the task level** — the model-rollout terminal pose error is flat over the orbit *to the float floor* (out-of-wedge / in-wedge ratio **1.000** on all 3 seeds — architectural) for the equivariant model, versus ×1.1–2.2 for the MLP under the *same* equivariant planner: the certificate now governs closed-loop *outcomes*, not just predictions; **(ii) it survives the real env** — executed on the real simulator the equivariant model's closed-loop block-angle error stays flat over the orbit to the measured precision (ratio **1.000** on all 3 seeds — the real interior physics is itself $\mathrm{SO}(2)$-equivariant to $\sim 10^{-5}$/step, so the *realized* outcome, not just the prediction, is orbit-invariant) while the MLP degrades ×1.6–3.6 out of the wedge; and **(iii) flatness is not bought by being worse** — in-wedge the equivariant model is *competitive*, its block-angle error (3.6–16° across seeds) bracketing the MLP's (8–18°), each better on some seeds, so the orbit-flatness is that of a usable controller (we claim no in-distribution win).

The mechanism is the cost landscape itself: the pose cost the planner optimizes is SO(2)-invariant, and under the equivariant model its value is orbit-invariant to the float floor (drift 1.1–1.5×10^{-7}), while under the MLP it is materially distorted (drift 0.19–0.31, ~ $10^6\times$ the equivariant floor). Two honesty notes, in the project's gating style. (a) That MLP cost-drift *straddles* a pre-registered absolute threshold (> 0.3, met on 1/3 seeds), so we report the eq/MLP *ratio* and the load-bearing orbit ratios — model-rollout ×1.000 and real-env ×1.000 versus the MLP's ×1.1–2.2 and ×1.6–3.6, on all 3 seeds — rather than the absolute drift, exactly as Experiment 9 reports its floor-penalty rather than the brittle `climb` sub-check. (b) Assumption (A5) genuinely *matters*: a scene-blind planner (raw, un-rotated CEM noise) keeps the equivariant model flatter on all 3 seeds (eq ×1.0–1.4 vs the MLP's ×1.5–3.5) but by a noisy margin — the *exact* guarantee needs the equivariant planner, while a realistic planner retains a softer version of the benefit. This is the certificate's strongest form — not a low error number but a **guarantee** that whatever competence holds in the wedge holds, zero-shot, over the entire orbit, to the float floor on every seed.

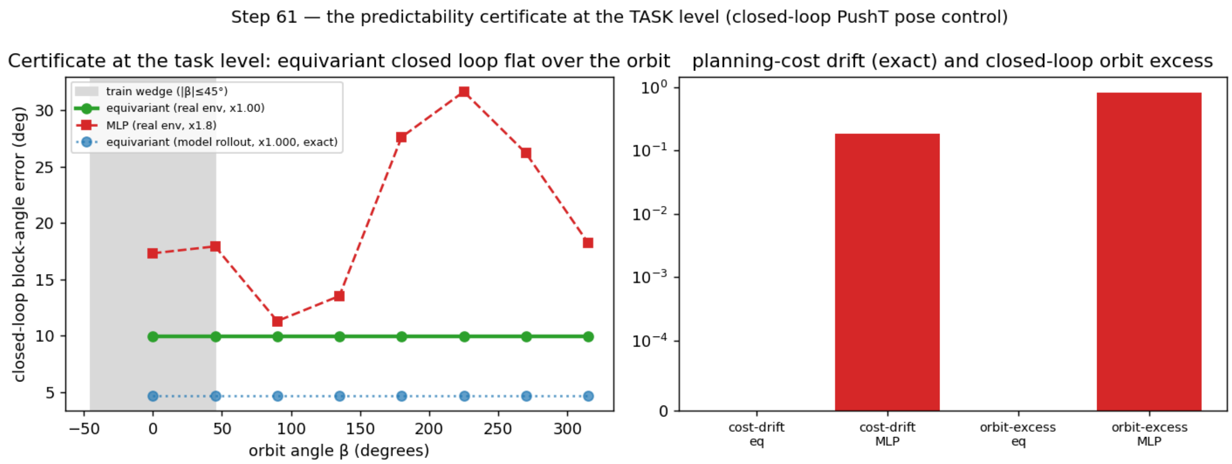

Figure 9: Experiment 11 (the certificate at the task level). Left: closed-loop block-angle error over the orbit of scene orientations — the equivariant model under a G-equivariant planner is exactly flat under model rollout (ratio 1.000) and flat on the real env, while a 4.3×-larger MLP degrades out of the training wedge. Right: the SO(2)-invariant planning cost is orbit-invariant to the float floor under the equivariant model and materially distorted under the MLP.

5.10 The certificate is not SO(2)-specific: a lift to the non-abelian SO(3)

Experiment 12 (the certificate on SO(3), 3D point clouds). Every result so far uses a *circle* — SO(2) or a discrete cyclic group. We lift the same multi-step rollout certificate to SO(3) **on 3D point clouds**, a genuinely larger, **non-commutative** group whose orbit is two-dimensional (an axis on S^2 times an angle). The dynamics is a constructed exactly-SO(3)-equivariant teacher on a 24-point cloud (drift + torque + anisotropic stretch) — like Experiments 1–7 a *toy* (the real-data anchor stays Experiment 9); the model is an e3nn point-cloud encoder whose latent is a stack of type-ℓ=1 vectors, with a jointly-equivariant Vector-Neuron predictor (the companion 3D line). We train one-step on a z-wedge of training rotations, then roll each model H=5 steps and measure rollout relMSE over the SO(3) orbit: the in-wedge identity, and out-of-distribution rotations (off-axis 90°, the antipode, random SO(3)).

Two findings, robust across 3 seeds (Figure 10). **(i) The certificate lifts to SO(3).** The learned equivariant model stays SO(3)-equivariant after training (residual ~ 10^{-5}) and its 5-step latent rollout is **exactly flat over the full SO(3) orbit** (out-of-wedge / in-wedge ratio **1.000**, all 3 seeds), where the MLP climbs ×2.1–5.7 out of the wedge — the configuration certificate is *not* an artifact of the abelian circle. **(ii) Structure versus scale, in 3D.** A 7.4× *smaller* equivariant model (17k vs 124k parameters) carries the certificate, while the larger MLP buys better in-wedge interpolation (relMSE 0.19–0.27 vs the equivariant model's 0.55–0.58) yet has none — the same *scale buys interpolation; structure buys a certificate* pattern as Experiment 7, now over SO(3).

The honest difference from SO(2) ("flat is not good", quantified). Unlike the real-PushT model of Experiment 9 — which was *competitive* in-distribution (0.13–0.15), so its flatness gave a clean out-of-distribution win — the small SO(3) point-cloud model's accuracy *floor* (≈ 0.57) is high enough that out of the wedge it is only **comparable** to the degraded MLP (whose far error 0.43–1.09 brackets that floor), not clearly below it. So the committed gate's `compete` sub-check (equivariant in-wedge error $< 1.5\times$ the MLP's) **fails**, and the run reports `INCONCLUSIVE` on it: we claim the lift of the *guarantee* to SO(3) and the structure-versus-scale pattern, **not** a 3D accuracy win — a competitive *and* flat 3D model needs more equivariant capacity than 1–2 GPUs afford. With Proposition 4 (angular momentum in the ℓ=1 block) and Experiment 6 (3D containment), this completes the 3D picture: representation theory *places* the SO(3) charges, and the certificate is *flat* over SO(3).

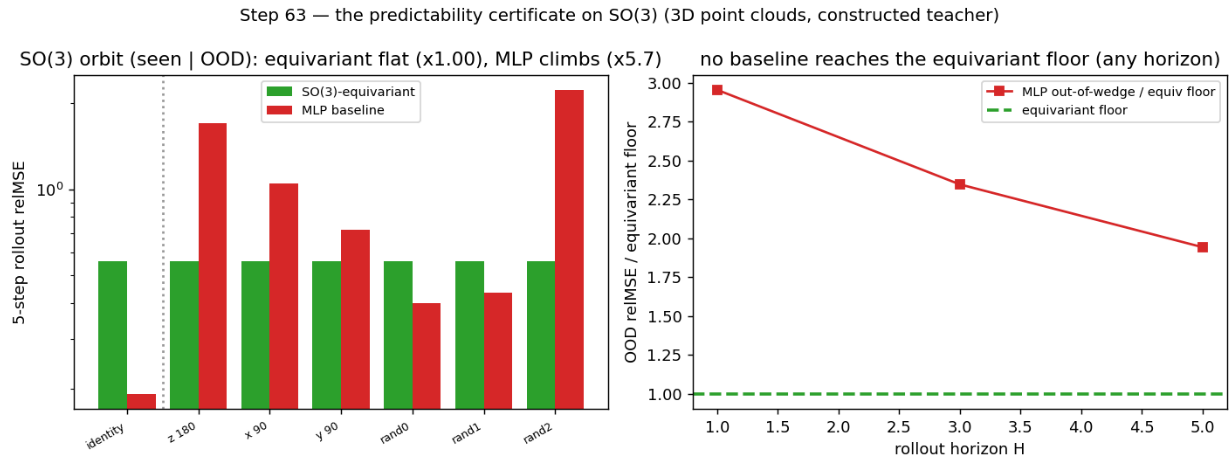

Figure 10: Experiment 12 (the certificate on SO(3), 3D point clouds, constructed teacher). Left: 5-step rollout relMSE over the SO(3) orbit (in-wedge identity | out-of-distribution rotations) — the learned equivariant model is exactly flat (ratio 1.000) while the 7.4×-larger MLP climbs out of the wedge. Right: across rollout horizon the MLP's out-of-wedge error stays at or above the equivariant floor, but the gap is modest because the small equivariant model's floor is high (the "flat is not good" caveat in 3D).

5.11 The certificate on raw pixels: frame averaging makes the prior accuracy-neutral

Experiment 13 (the certificate on rendered pixels, and what the prior costs). Every experiment so far reads a *structured* state or point cloud. We close the modality gap on the hardest input — **rendered RGB frames** of PushT (`experiments/step64`) — over the exact C_4 subgroup (a 90° scene rotation is a bit-exact pixel permutation on the odd-sized grid; continuous SO(2) on a pixel grid is interpolation-floored, so C_4 is the grid-exact group). Theorem A already guarantees the certificate *transfers* for any exactly equivariant (E, f); the real question is whether the equivariance prior **costs accuracy** — the standing pixel limitation of earlier drafts, where an e2cnn-steerable pixel JEPA underfit badly. We compare three latent world models trained identically (EMA-target JEPA + VICReg) on the same frames: the C_4-steerable e2cnn model (the incumbent); an unconstrained CNN/MLP (the accuracy reference, no flatness guarantee); and a **frame-averaged** model — a *plain* CNN encoder and MLP predictor made *exactly* C_4-equivariant by a Reynolds average over the four grid rotations with a ρ-correction, $E(o) = \frac{1}{4}\sum_k \rho(g_k)^{-1}\phi(g_k \cdot o)$ (Puny et al., 2022), with $\rho = I \oplus (R_k \otimes I)$ orthogonal so Theorem A applies. Accuracy is read collapse-robustly as the fraction of *centered* latent variance unexplained (FVU; $< 1 \iff$ beating predict-the-mean), since an uncentered relMSE is deflated by a large constant latent mean. Three findings (3 seeds; `tests/test_step64.py` certifies E, f equivariant to $\sim 10^{-7}$ at init **and** after training):

- **(a) Flatness transfers exactly.** The frame-averaged rollout is flat over the C_4 orbit to the float floor (ratio 1.000, every seed), with E, f equivariant to $\sim 10^{-7}$ — the certificate is not an artifact of structured state.
- **(b) Frame averaging makes the prior accuracy-neutral.** The frame-averaged model **matches or beats the unconstrained CNN** on FVU (ratio 0.68–1.07, mean 0.84 over seeds) with a *healthier* latent (participation ratio 2.8–4.3 vs. the CNN's 2.2–2.6) — so the earlier steerable underfit was an e2cnn *optimization* artifact, **not** a cost of equivariance. Over the rollout horizon the contrast sharpens: the steerable model **diverges** (its FVU grows 160–1600× its one-step value), while the frame-averaged model stays **stable** ($\leq 1.2\times$, usually below the CNN's 1.1–1.6×) — equivariance done right is reliable, not just flat.

- **(c) The residual is not equivariance.** Absolutely, *no* pixel model beats predict-the-mean here (FVU $>$ 1), but this is **architecture-agnostic** and not a measurement artifact: even the unconstrained CNN's *fair* one-step accuracy against its own EMA target is FVU $>$ 1 (1.8–2.2), as is the frame-averaged model's (1.7–2.0). The cause is the JEPA latent itself — VICReg forces unit variance on all D=64 dimensions, but PushT dynamics is low-dimensional (latent rank $\approx$ 3), so most dimensions carry unpredictable anti-collapse variance. A strong few-step pixel-latent predictor at 1-GPU scale is the residual open problem, shared by *every* architecture; the certificate (flatness), the accuracy-neutrality of the prior, and the rollout stability hold regardless.

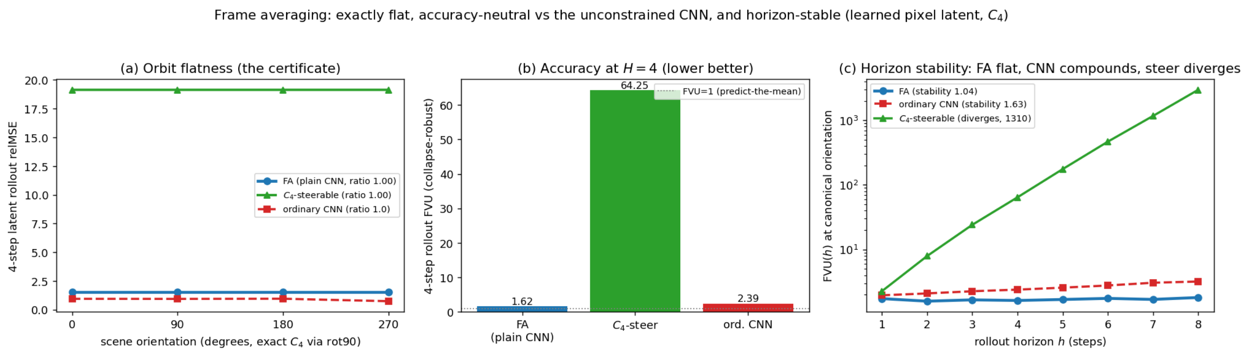

Figure 11: Experiment 13 (the certificate on rendered pixels, C_4). **(a)** 4-step latent-rollout relMSE over the orbit of scene orientations: the frame-averaged model is flat to the float floor (ratio 1.000); the ordinary CNN is also orbit-flat (PushT's pixel stream is approximately C_4-symmetric, the augmentation regime of §5.8). **(b)** Collapse-robust accuracy (FVU) at the canonical orientation: frame averaging matches/beats the unconstrained CNN and is far better than the steerable incumbent (dotted line = predict-the-mean). **(c)** FVU vs. rollout horizon (log scale): the steerable rollout *diverges* while the frame-averaged model stays *stable*. Absolute FVU $>$ 1 for all models is an architecture-agnostic JEPA-latent property, not an equivariance cost.

The pixel takeaway is therefore sharper than "structure underfits": structure transfers the certificate *exactly and for free* — frame averaging is flat, accuracy-neutral, and horizon-stable — and what remains open, a strong few-step pixel predictor at small scale, is not an equivariance problem at all.

5.12 The horizon law on a learned model of *real chaotic* dynamics (Experiment 14)

Every horizon result so far lives on a *constructed* spectrum: §5.2's staircase uses a synthetic latent with planted multipliers, and `step65` instantiates Proposition 6 analytically. The decisive test of Proposition 7(a) is whether the $T(\epsilon) \sim \log(1/\epsilon)/\lambda_1$ law lifts to a *learned* model of a system that is *genuinely chaotic* — a real positive Lyapunov exponent, not a planted one. We take the **Lorenz attractor** (σ=10, ρ=28, β=8/3; singular-hyperbolic [Tucker 2002], $\lambda_1 \approx 0.9056$, with the $\mathbb{Z}_2$ symmetry $(x, y, z) \mapsto (-x, -y, z)$), integrate it with RK4, train a plain one-step MLP of the Δt map, and run the §5.2 certified-horizon staircase **on the learned model**.

The law lifts cleanly (3 seeds, all pass). The one-step map is learned to relMSE $< 10^{-4}$; the certified horizon $T(\epsilon_0)$ is **linear in** $\log(1/\epsilon_0)$ with R^2=0.975/0.995/0.990; and the staircase slope measures the *learned model's* Lyapunov exponent $\hat{\lambda}_1 = 1/(\text{slope} \cdot dt) = 0.895/0.919/0.977$ — reading the slope off the model's own first-crossing growth is definitional, so the load-bearing claim is the **agreement with the true integrator**: $\hat{\lambda}_1$ matches textbook λ_1=0.9056 to a relative error of 1.2%/1.4%/7.9%. A model can be one-step-accurate (relMSE $< 10^{-4}$) yet drift to a wrong *multi-step* rate; that it does not is the content. (We use the staircase slope, not an early-window finite-time exponent: the latter is window-sensitive on Lorenz — on one seed the early window caught a transient super-expansion $\hat{\lambda} \approx 3.7$ — whereas the staircase, fit over the whole horizon range, is stable across seeds. A unit test validates the staircase protocol on the *true* integrator: Lorenz $\mathbb{Z}_2$-equivariance holds to machine precision and the slope recovers λ_1 to 2.1%, so the law is not an artifact of any one trained model.) This is Proposition 7's branch (a) in action: the certificate's horizon axis is real-world-meaningful precisely because the dynamics is spectrally non-degenerate — and the lift is the *empirical* statement (the learned model preserves λ_1), since classical shadowing does not formally apply to singular-hyperbolic Lorenz. The contrast with the PushT interior — branch (b), $\lambda_1 \approx 0$, where a learned-model horizon probe finds the local spectrum does *not* predict the rollout (R^2=0.02) — is exactly the dichotomy Proposition 7 predicts, not a contradiction: the horizon certificate is informative on chaotic dynamics and vacuous on near-neutral dynamics, and one can tell which regime one is in by measuring λ_1.

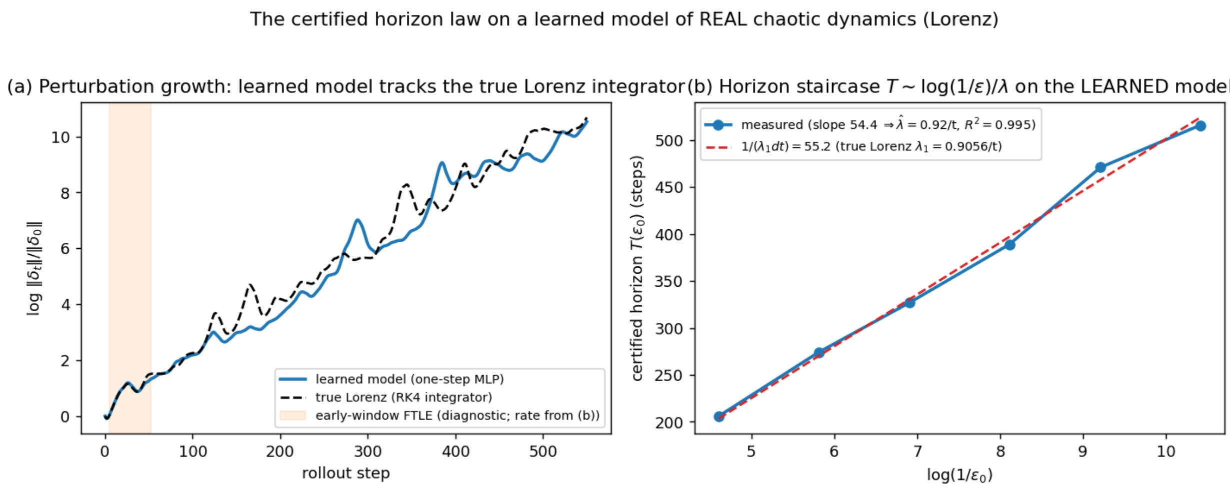

Figure 12: Experiment 14 (the certified-horizon law on a learned model of real chaotic dynamics, Lorenz). **(a)** Perturbation growth: the learned one-step model (blue) tracks the true Lorenz integrator (black dashed) over 550 steps. **(b)** The certified horizon $T(\epsilon_0)$ on the *learned* model is linear in $\log(1/\epsilon_0)$ (R^2=0.995, seed 1), and the measured slope sits on the prediction $1/(\lambda_1 dt)$ from the textbook Lorenz exponent — the $T \sim \log(1/\epsilon)/\lambda$ law of Theorem B, lifted to a learned model of a genuinely chaotic system (Proposition 7(a)).

5.13 The horizon lift is a property of *chaotic dynamics*, not of Lorenz (Experiment 15)

One system is an existence proof; a *law* needs a class. We run the identical learned-model protocol on three chaotic systems spanning an order of magnitude in exponent and two dynamical kinds: the **Hénon map** (a=1.4, b=0.3; a discrete 2D map, documented $\lambda_1 \approx 0.419$/step), the **Rössler flow** (a=b=0.2, c=5.7; a continuous 3D ODE with a *small* exponent $\lambda_1 \approx 0.0714$/t — the stress case, since its horizon is long), and **Lorenz** as the anchor. For each we train a plain one-step MLP on on-attractor data and read $\hat{\lambda}_1$ off the certified-horizon staircase. **The law lifts across the class** (3 seeds; aggregate in `step71_multichaos_horizon_seeds.json`): the staircase is linear (R^2=0.96–1.00) and its slope recovers the textbook exponent — Hénon $\hat{\lambda}_1$=0.45–0.47 (rel-err 8–12%, 3/3 **seeds**), Lorenz 0.90–0.95 (1–5%, 3/3), and the small-exponent Rössler 0.065–0.066 (8–9%, 2/3 — on one seed its ~1500-step horizon failed to cross enough resolutions for a clean fit, the honest fragility of recovering a *tiny* exponent over a long horizon). Every *seed* passes overall (Hénon and Lorenz always carry the ≥2-of-3 gate). Crucially, Rössler — whose tiny λ_1 demands a ~ 1500-step horizon — illustrates **Proposition 8's $O(\delta)$ model-fidelity bias directly**: under *reduced* training its recovered exponent overshoots to ~44%, and it falls to ~8% as the one-step error δ drops with fuller training, exactly the "better fidelity ⇒ tighter exponent" the bound predicts. A unit test isolates the *other* bias term: on the *true* maps/flows (no learned model at all) the staircase still recovers λ_1 only to ~9–10%, so a chunk of the residual is the finite-horizon truncation $|\lambda_{1,T} - \lambda_1|$ of Proposition 8, not model error. So the horizon axis is not a Lorenz coincidence: **wherever the dynamics is genuinely chaotic, a one-step-accurate learned model's certified-horizon staircase recovers the true chaos rate**, with a bias the finite-horizon continuity bound (Proposition 8) both predicts and decomposes.

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6. Related Work

The paper sits among a cluster of concurrent *structure-for-prediction* works. The table places each against the certificate's three axes and the *kind* of guarantee it offers (✓ provides; ~ partial/empirical; — not addressed); the prose elaborates below.

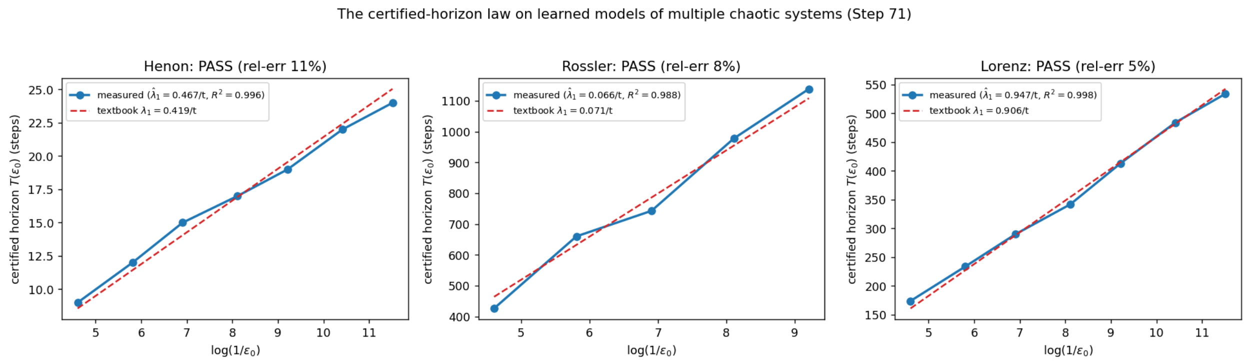

Figure 13: Experiment 15 (the certified-horizon law across a class of learned chaotic models). The horizon staircase $T(\epsilon_0)$ on the learned model of each system is linear in $\log(1/\epsilon_0)$ and its slope (blue $\Rightarrow \hat{\lambda}_1$) sits on the textbook exponent (red dashed): a 2D map (Hénon), a small-exponent flow (Rössler), and a large-exponent flow (Lorenz). The law is a property of chaotic dynamics, not of Lorenz.

Work	Core mechanism	Config. $\langle S \rangle$	Horizon$\times\epsilon$	Closed-loop	Guarantee kind
BRo-JEPA	$\mathbb{Z}/10$ cyclic predictor	~ one group	—	—	empirical zero-shot
UWM-JEPA	$U(d)$ unitary predictor	~ one group	—	—	empirical
UR-JEPA / LeJEPA	latent-geometry prior (an/isotropy)	—	—	—	distributional (2nd-order)
IMWM	oracle-bypass; residual = search	—	—	~ search budget	diagnostic
LDA	SE(3)-intrinsic diffusion (action side)	—	—	~ policy	empirical, mod-est+robust
Symmetry–Data rate	aug+TTA = equivariance; wrong-group harmful	~ one group	—	—	empirical scaling law
companion line	exact flatness: 1 element, 1 resolution	~ one element	—	✓ invariance	proved (a corner)
certified equivariance (robustness / conformal)	orbit-margin / group-averaged conformal score	~ one group	— (single-shot)	—	proved, but **single-shot**, forward-only
Noether Networks / Razor	meta-learn conserved quantities	—	~ better avg prediction	—	average accuracy, no guarantee

Work	Core mechanism	Config. $\langle S \rangle$	Horizon$\times\epsilon$	Closed-loop	Guarantee kind
this paper	equivariant **predictability certificate**	✓ $k \Rightarrow \langle S \rangle$ (Thm A, Lem 1)	✓ $T_j(\epsilon) \sim \frac{\log(1/\epsilon)}{\lambda_j}$, **tight** (Thm B, Prop 6)	✓ exact (Exp 11)	**a-priori, per-situation, computable (Alg 1)**

The throughline: prior work supplies *mechanisms* (equivariant predictors), *priors* (latent geometry), or *diagnostics* (oracle-bypass) — each empirical or distributional and on a single axis; this paper supplies a *provable, computable region* across configuration × horizon × resolution, with the Noether hinge (§4) tying the axes together.

Equivariance and constant error (our companion line). A companion paper establishes, for a single group element and one resolution, that exact equivariance makes one-step error constant across the group, with exact-flatness and closed-loop-invariance corollaries. The present paper lifts that corner along three axes: Theorem A is its multi-step, full-monoid generalization; Theorem B adds the horizon × resolution stratification; and Lemma 1 turns k generator checks into an exponential certified set.

Certified equivariance is single-shot; ours is multi-step, with a converse. The closest *guarantee-bearing* relatives live in robustness and conformal prediction, not world models, and a careful comparison sharpens exactly what is new here. Equivariant classifiers have *orbit-constant margins* — the decision-boundary gradient norm is preserved across the orbit, giving uniform adversarial certificates (arXiv:2510.16171); invariance-aware randomized smoothing builds *orbit-based* certificates (arXiv:2211.14207); and Equivariantized Conformal Prediction (eCP, arXiv:2602.03986) *group-averages the nonconformity score* — frame averaging for distribution-free coverage — to tighten prediction sets over an orbit. Three differences are load-bearing. **(i) Time.** All of these are *single-shot*: a classifier margin or a one-shot prediction set, with the guarantee attached to one input, never *propagated through an H-step latent rollout*; none has a prediction-*horizon* axis, a predictor *spectrum*, or a *conservation* law. Our certificate is precisely a multi-step statement, stratified by the Lyapunov spectrum (Theorem B, made *tight* by Proposition 6) and extended to all horizons on conserved channels (Proposition 5). **(ii) Direction.** This literature proves *equivariance ⇒ orbit-constant guarantee* — at root the classical fact that an equivariant estimator's risk is constant on orbits, specialized to margins or coverage; it proves no converse. Lemma 2 supplies the **converse** (orbit-constant error $\iff$ equivariance), which is what makes "no non-equivariant model has the certificate at any scale" a theorem rather than an observation. **(iii) Object.** Frame averaging unifies the two literatures concretely but at different layers: eCP averages a *score* post-hoc for coverage; we average an *encoder/predictor* (Experiment 13) for an in-model multi-step certificate — the same Reynolds operator, opposite ends of the pipeline. Separately, **learned-conservation** methods — Noether Networks (Alet et al., 2021; arXiv:2112.03321) and Noether's Razor (2024; arXiv:2410.08087) — meta-learn conserved quantities to *improve average prediction* by shrinking the hypothesis space; our Noether hinge instead *certifies* (Proposition 5 turns a conserved charge into an a-priori long-horizon guarantee, Proposition 4 says which isotypic block must carry it) — guarantee versus average accuracy. Finally, **Jacobian-regularized world models** (arXiv:2501.00195) damp rollout error propagation by penalizing the transition Jacobian — a heuristic; Theorem B is the *provable* version of the same intuition, reading a per-channel certified horizon $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$ off the spectrum rather than regularizing toward stability, and (Proposition 6) characterizing how that horizon shrinks when the symmetry is only approximate.

Equivariant predictors. Block-rotation predictors (BRo-JEPA, arXiv:2606.01372) match a $\mathbb{Z}/10\mathbb{Z}$ cyclic structure to the latent — "add k" becomes "rotate $k\theta$" — and report 99.46% best-config zero-shot transfer on MNIST modular arithmetic; unitary-predictor world models (UWM-JEPA, arXiv:2605.25313) do the analogous thing with $U(d)$. Theorem A explains *why* such zero-shot transfer occurs (the representation cancels inside the norm — orthogonal for the former, unitary for the latter), and the closure lemma quantifies *how far* it reaches (the whole generated monoid from its generators). These works share our toy-construction caveat, and we cite them as cross-group mechanism corroboration rather than scale results.

Latent-geometry regularizers. LeJEPA (Balestriero and LeCun, 2025) pushes the latent toward an isotropic Gaussian; UR-JEPA (Le, 2026; arXiv:2606.01443) challenges that target, arguing isotropy conflicts with the manifold hypothesis and that a data-discovered anisotropy is preferable. Our latent's anisotropy is neither isotropic nor data-discovered but **group-prescribed**: the covariance is pinned to the representation, $\rho\Sigma\rho^\top = \Sigma$ (measured residual 3×10^{-4}, versus

1.04 for a non-equivariant baseline). In a head-to-head (companion line, single seed) the data-fit anisotropy is best *in-distribution*, but only the group-prescribed anisotropy transfers out of distribution (a ~ 320× gap, consistent with the three-seed 68–320× and 10–155× of Experiments 8 and 7). The certificate is therefore complementary — a first-order, per-situation guarantee from the group, not a second-order distributional prior — and it predicts precisely *when* a discovered anisotropy will fail to generalize (off the training orbit).

Oracle-bypass diagnostics. IMWM (Gao et al., 2026; arXiv:2606.01626) uses the same falsifiable move as our capacity-ladder experiments — replace the learned model with oracle dynamics and see what is still missing — and finds a finite-budget planner *still* fails. It attributes the residual to **search** (proposal-sampling volume), whereas our analogous diagnosis attributes it to **representation** (permutation-invariant encoder pooling). These are complementary bottlenecks: a predictability certificate bounds the *model's* error over $\langle S\rangle \times T \times \epsilon$, and makes no claim about the planner's search budget downstream.

Geometry on the action side. LDA ("The Lie We Tell", Chuang et al., 2026; arXiv:2606.01847) names the *Euclidean Fallacy* — flattening an SE(3) pose into $\mathbb{R}^{12}$ breaks the manifold constraint, the coordinate-change equivariance, and geodesic optimality — and fixes it by running diffusion *on* SE(3) (tangent-space score, exponential-map retraction). It states our core motivation — *do not flatten geometric quantities* — on the action side. As a diffusion policy its gains are modest but robust (CALVIN average task length 3.27 → 3.51, +7.3%), the same profile as equivariant methods generally; our framing explains why a geometric prior buys a *kind* of guarantee (consistency across the group, robustness out of distribution) rather than a uniform accuracy jump.

Concurrent corroboration: the symmetry–data exchange rate, and learned irreps. Two concurrent works sharpen pieces of our picture. *Measuring the Symmetry–Data Exchange Rate* (Adly, 2026; arXiv:2606.01090), on a controlled C_n task, independently reports the two facts our §5.8 and §5.5 rest on: (i) a non-equivariant model with augmentation **plus test-time orbit averaging matches the equivariant model essentially exactly** (coincident per-epoch validation curves) — the single-orbit tie of our augmentation experiment (Experiment 10); and (ii) a **misaligned (wrong-group) symmetry constraint is *actively harmful*** (95% CI excluding zero), not merely unhelpful — the approximate-symmetry threshold of our §5.5. It frames the trade as a "symmetry–data exchange rate," the average-case dual of our worst-case separation (§3.3: structure certifies the ϵ-independent orbit, data only an ϵ/L tube); like ours it carries the controlled-toy caveat, and it stops short of the multi-axis certificate, the spectral horizon law, the closed-loop clause, and the Noether hinge. On the learning-dynamics side, *Neural Networks Provably Learn Spectral Representations for Group Composition* (He et al., 2026; arXiv:2606.02993) proves that gradient flow on a group-composition task drives each neuron to a single **irreducible representation** — a provable account of *how* the isotypic structure our Proposition 4 places, and our §5.6 discovery experiments recover, emerges under training.

Predictability horizons. The $T(\epsilon) \sim \log(1/\epsilon)/\lambda$ law is classical for dynamical systems (Lyapunov; numerical weather prediction), and that the *local* spectrum governs the *asymptotic* rate is the content of the Oseledets multiplicative ergodic theorem; on *uniformly* hyperbolic systems the shadowing lemma (Anosov–Bowen; Pilyugin) further bounds the forecast-horizon *floor* of a perturbed (learned) model — though it controls orbit error, not the exponent, and does not formally cover singular-hyperbolic Lorenz (Tucker 2002). Our contribution is to (i) measure the law on a *learned latent world model*, including a learned model of genuinely chaotic dynamics (Lorenz, Experiment 14: the learned model's Lyapunov exponent, read as the staircase slope, *matches* the true λ_1 to 1–8% — an empirical lift, since shadowing does not transfer exponents); (ii) characterize *when* the local spectrum certifies the horizon — Proposition 7's dichotomy: informative on spectrally non-degenerate dynamics ($\lambda_1 > 0$), vacuous on near-neutral dynamics ($\lambda_1 \approx 0$, the PushT interior) — synthesizing Oseledets (for the rate) and shadowing (for the floor) with the certificate; (iii) tie its slow subspace to the group-invariant subspace via the Noether hinge; and fold all of it into a single multi-axis certificate for equivariant models.

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7. Limitations

- **Scale and scope.** All experiments are CPU/1-GPU. Experiments 9 and 11 validate the central claim on a real physics-engine contact simulator (PushT) — dynamics we did not author — at the *prediction* level (Experiment 9) and the closed-loop *task* level (Experiment 11, where the certificate becomes orbit-invariant pose control: out of the training wedge the equivariant controller stays flat while a scaled baseline degrades, the two being *comparable* in-distribution — we claim no in-distribution win). This is still *structured state* (not pixels) and a single SO(2) group; the out-of-distribution *prediction* gap is *modest* (2–4×, large only because a single PushT step is easy to predict in-distribution); and the closed-loop *out-of-wedge* advantage is shown on the contact-dominated pose task specifically — a position-only push stays a tie, since a near-linear agent subsystem carries it out of distribution. We claim a mechanism and a tool, demonstrated cleanly, not a scaled benchmark.

- **Pixels (Experiment 13, §5.11): structure is free; absolute accuracy is the open part.** The certificate transfers *exactly* to rendered pixels, and — via frame averaging — at **no accuracy cost** relative to an unconstrained CNN (matches-or-beats it on collapse-robust FVU, with a healthier latent and a horizon-stable rollout; §5.11). The honest limitation is *absolute*, not comparative: at 1-GPU scale **no** pixel model, equivariant or not, beats predict-the-mean at a multi-step horizon (FVU >1 even for the unconstrained CNN's fair one-step accuracy). This is an architecture-*agnostic* property of the JEPA latent — VICReg's anti-collapse variance on low-dimensional dynamics — not a cost of the prior. A strong few-step pixel-latent predictor at small scale remains open; everything the certificate promises (exact flatness) holds regardless.
- **Scope of the exact certificate.** Theorem A requires (A3): the group must be a symmetry of the *dynamics*, not merely the encoder. The exact certificate therefore holds where the group is a genuine dynamical symmetry (orbital and conservative systems, free space, idealized manipulation). Everywhere else one is in Theorem B's approximate regime, whose degradation we *measure* rather than assume: Experiment 8 shows it is graceful ($\propto \epsilon_{\text{world}}$) up to a measured threshold $\epsilon_{\text{world}} \approx 0.01$–$0.06$ (seed-dependent; one seed crosses over as early as $\epsilon \approx 0.008$). The honest reading is "exact on symmetric dynamics; gracefully approximate, with a measured boundary, elsewhere."
- **Theorem B's horizon is tight, lifts to learned chaotic dynamics, and has a characterized scope; its constants are not estimated.** The certified horizon $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$ is bounded on *both* sides — Theorem B and the matching lower bound of Proposition 6 — so the horizon's *form* is tight (§5.2 recovers it to 0.4%). It is no longer only a synthetic-spectrum result: Experiment 14 lifts it to a *learned* model of genuinely chaotic dynamics (Lorenz), where the learned model's Lyapunov exponent (the staircase slope) *matches* the true λ_1 to 1–8% (R^2=0.975–0.995). Proposition 7 characterizes *when* the local spectrum certifies the horizon — informative on spectrally non-degenerate dynamics ($\lambda_1 > 0$; Oseledets gives the rate rigorously), vacuous on near-neutral dynamics ($\lambda_1 \approx 0$, the PushT interior, where a learned-model probe gives R^2=0.02) — so the PushT horizon negative is *explained*, not a gap. What remains un-tightened: the prefactor (the constants c_j are not estimated); the *isotypic* refinement (channels by ℓ-type) is asserted, not separately measured (Experiment 3 measures the scalar law, Experiment 1 splits contractive from expansive but not by ℓ-type); and the *lift* of the rate to a learned model is, for Lorenz, **empirical** — shadowing bounds only the forecast-horizon floor and does not transfer the exponent, and classical shadowing does not formally cover singular-hyperbolic Lorenz; that the learned model preserves λ_1 (verified only via one-step error, an L^2 proxy for C^1-closeness) is the experimental finding, not a theorem.
- **The Noether hinge's forward direction is proved; its hypotheses are measured.** The *placement* of each conserved charge by isotypic type (Proposition 4 — energy in ℓ=0, angular momentum in the ℓ=1 block via the unique degree-2 cross product) is forced by representation theory, and the *conserved* ⇒ *slow* direction is now a theorem (Proposition 5: a charge conserved to one-step defect η has prediction error $\leq T\eta$, linear not exponential — certified to all horizons at η=0). What remains *assumed or measured*: that the learned latent flow is Hamiltonian with a G-invariant Hamiltonian (so the symmetry is a *dynamical* symmetry — the encoder essentially symplectic); the value of the defect η (exact only for the momentum map under a G-equivariant symplectic discretization, $O(\Delta t^p)$ for energy, measured for a generic learned f); and the **non-converse** (slow $\not\Rightarrow$ conserved). Proposition 5 is moreover a statement about the *charge value*, not full-state prediction on the conserved subspace. Its lift to a *real, contact-rich embodied* model — approximate symmetry, latent learned end-to-end — is the primary open problem; the 3D contact experiment is a clean two-body toy whose clean-containment gate is `INCONCLUSIVE` (resolved in type-and-degree form by Experiment 6).
- **§5.6 re-frames companion-line results**, not fresh runs: the decoder-free reach is *exact transfer* (unseen/seen ratio 1.000) at a *partial* absolute success ($\sim$ 0.59 of the orientation gap). We report the transfer ratio as the load-bearing claim, not absolute task success.
- **"Flat" is not "good."** A constant error across the group says nothing about its magnitude; Theorem A certifies *consistency*, and the "scale never reaches the certificate" claim is an *architectural* fact, **proved** in Lemma 2 (orbit-constant error against every equivariant target $\iff$ equivariance) and quantified in §3.3 (a non-equivariant L-Lipschitz learner certifies only an ϵ/L-tube around the data) — not the conclusion of a finite sweep.

Falsifiability. The framing makes refutable predictions, not after-the-fact rationalizations. A finite non-equivariant network attaining exact orbit-flatness would break Theorem A's architectural premise; a symmetric-dynamics system whose conserved/slow modes lay *outside* the invariant ⊕ conserved-equivariant subspace would break the hinge; and a channel whose certified horizon failed to scale as $\log(1/\epsilon)/\lambda_j$ would break Theorem B. We have not observed these; they are the experiments that would sink the thesis.

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8. Conclusion

For equivariant world models, structure delivers a *kind* of result scaling cannot: a **predictability certificate** — a provable, computable, training-free region across configuration, horizon, and resolution. We established the master theorem (Theorem A, the exact configuration certificate, proved — its closed-loop clause under the assumed equivariant-planner condition; Lemma 2, its converse, characterizing the certificate *as* equivariance; Theorem B, the spectral horizon × resolution law, stated as a bound and measured), exhibited the certified region as the coarse-invariant-slow-low-composition corner, proved the Noether hinge's forward direction that unifies the symmetry and time axes (Proposition 5: conserved ⇒ slow) and measured its defect, and showed — empirically and via the quantitative separation of §3.3 — that an 88×-scaled non-equivariant model buys interpolation but never the certificate, an architectural impossibility rather than an unfinished sweep. *Scale buys interpolation; structure buys a certificate.* The same criterion that tells you which compositions a robot policy will handle zero-shot also tells you why eclipses are forecastable for millennia while weather is not — one structural law, read across domains.

References

Concurrent and prior work referenced above (arXiv identifiers from the project's source notes; external numbers are quoted from the cited works):

- **BRo-JEPA** — block-rotation $\mathbb{Z}/10\mathbb{Z}$ predictor; 99.46% best-config zero-shot on MNIST modular arithmetic. arXiv:2606.01372.
- **UWM-JEPA** — unitary-predictor ($U(d)$) belief-space world model. arXiv:2605.25313.
- **UR-JEPA** (Le, 2026) — uniform-rectifiability latent regularizer; manifold-hypothesis critique of isotropic SIGReg. arXiv:2606.01443.
- **IMWM** (Gao et al., 2026) — oracle-bypass diagnosis localizing the residual to planner *search*. arXiv:2606.01626.
- **LDA / "The Lie We Tell"** (Chuang et al., 2026) — the *Euclidean Fallacy*; SE(3)-intrinsic diffusion policy; CALVIN average task length 3.27 → 3.51 (+7.3%). arXiv:2606.01847.
- **LeJEPA** (Balestriero and LeCun, 2025) — SIGReg / isotropic-Gaussian self-supervised objective (the latent-geometry target that UR-JEPA, and our group-prescribed anisotropy, depart from).
- **Symmetry–Data Exchange Rate** (Adly, 2026) — controlled C_n scaling-law study; augmentation + test-time orbit averaging matches equivariance, and a wrong-group constraint is actively harmful — the average-case dual of our §3.3 separation, corroborating Experiment 10 (§5.8) and §5.5. arXiv:2606.01090.
- **Spectral Representations for Group Composition** (He et al., 2026) — gradient flow on a group-composition task provably drives neurons to single irreducible representations; the learning-dynamics complement to Proposition 4 and the §5.6 discovery experiments. arXiv:2606.02993.
- **Frame Averaging** (Puny et al., 2022) — a Reynolds-operator construction that makes an arbitrary backbone exactly invariant/equivariant by averaging over a group frame; the method we use in `experiments/step64` to obtain an exactly C_4-equivariant *pixel* encoder/predictor from plain `torch` nets (so the §7 pixel certificate is flat *and* accuracy-neutral relative to the unconstrained CNN). arXiv:2110.03336.

Certified equivariance, conservation, and stability (the §6 carve-out — these are the guarantee-bearing neighbors; all single-shot and forward-only, none with the multi-step horizon, the converse, or the Noether tie):

- **Orbit-constant margins for equivariant networks** — equivariant classifiers have an orbit-invariant margin / decision-boundary gradient norm, giving uniform adversarial certificates across the orbit; single-shot classification, forward direction only. arXiv:2510.16171.

- **Equivariantized Conformal Prediction (eCP)** — group-averages the nonconformity score (frame averaging for distribution-free coverage) to tighten orbit-wide prediction sets; single-shot, per-sample, no horizon. arXiv:2602.03986.
- **Invariance-aware randomized smoothing** — orbit-based robustness certificates combining invariance with smoothing. arXiv:2211.14207.
- **Noether Networks** (Alet et al., 2021) — meta-learn conserved quantities inside the prediction loop to improve *average* prediction (hypothesis-space shrinkage); no a-priori guarantee. arXiv:2112.03321.
- **Noether's Razor** (2024) — learns symmetries-as-conserved-quantities by Bayesian model selection with an Occam effect; average accuracy, not a certificate. arXiv:2410.08087.
- **Jacobian-regularized world models** ("Towards Unraveling and Improving Generalization in World Models", 2025) — penalizes the latent-transition Jacobian to damp rollout error propagation; the heuristic version of Theorem B's provable per-channel horizon. arXiv:2501.00195.

Classical dynamical-systems results behind the horizon axis (Theorem B, Proposition 7):

- **Oseledets multiplicative ergodic theorem** (Oseledets, 1968) — under an ergodic invariant measure with $\log^{+}\|D\phi\| \in L^1$, the finite-time Lyapunov exponents converge a.e. to constants along the Oseledets filtration; this is what licenses using a *locally/finite-time*-measured top exponent as the asymptotic rate λ_1 in Proposition 7's rate half.
- **Shadowing lemma** (Anosov; Bowen; Pilyugin, *Shadowing in Dynamical Systems*, 1999) — on uniformly hyperbolic sets, pseudo-orbits are shadowed by true orbits; bounds the *forecast-horizon floor* of a perturbed (learned) model, but controls orbit error, **not** Lyapunov exponents (which are only upper-semicontinuous under C^1 perturbation).
- **Continuity of Lyapunov exponents under domination** (Bochi & Viana, *The Lyapunov exponents of generic volume-preserving and symplectic maps*, Ann. of Math. 2005) — Lyapunov exponents are continuous on the domain of *dominated splitting*; this is the asymptotic face of Proposition 8's finite-horizon exponent-transfer bound (the finite-T version is elementary continuity of a matrix product, the T-uniform constant needs domination).
- **Lorenz attractor is singular-hyperbolic with an SRB measure** (Tucker, 2002, *A rigorous ODE solver and Smale's 14th problem*; Araújo–Pacífico–Pujals–Viana, 2009) — gives the ergodic measure for the MET on Lorenz, but is *not* uniformly hyperbolic, so classical shadowing does not formally apply near the singularity (hence Experiment 14's exponent-lift is empirical, not theorem-backed).

The $T(\epsilon) \sim \log(1/\epsilon)/\lambda$ predictability-horizon law itself is classical (Lyapunov / Lorenz; standard numerical-weather-prediction practice).

Appendix A. Reproducibility

Every experiment sets random seeds explicitly, prints an `INCONCLUSIVE` verdict rather than loosen a gate, and writes its figure and JSON to `papers/figures/`. The multi-seed steps commit per-seed JSONs (`papers/figures/step5*_seeds.json`, `step59_pusht_certificate_seeds.json`, `step60_augmentation_seeds.json`, and `step61_closed_loop_certificate_seeds.json`, `step63_se3_certificate_seeds.json`, `step64_frame_averaged_pixel_seeds.json`, `step70_lorenz_horizon_seeds.json`, and `step71_multichaos_horizon_seeds.json`, seeds 0/1/2), regenerated by `experiments/aggregate_seeds.py`; every range quoted above is the seed min–max from those files. The configuration-flatness experiment (Experiment 1) self-aggregates its three seeds into the means in `step47_certificate.json`. The full test suite passes together; `tests/conftest.py` isolates the float64 experiments from the float32 codebase. One command reproduces the paper end-to-end — `make paper2` (multi-seed re-runs → figures → tests → PDF), with `make paper2-quick` for the figures-and-PDF fast path; everything is CPU/MPS, no CUDA.

Exp.	Result	Code	Test	Seeds	Headline number
1	Exact compositional flatness + spectrum	`experiments/step47_certificate.py`	`tests/test_step47_certificate.py`	3	relMSE ×1.00 flat (m=0..8); 48/96 contractive
2	$\mathbb{Z}_2^6$ exponential certificate	`experiments/step49_iching_certificate.py`		1	6 generators → all 64; worst relMSE ~ 10^{-33}
3	Horizon × resolution staircase	`experiments/step52_horizon_resolution.py`	`tests/test_step52_horizon_resolution.py`	3	$\hat{\lambda}$=0.69 vs $\ln 2$; slope ≈ $1/\lambda$
4	Noether hinge (2D)	`experiments/step50_noether_hinge.py`	`tests/test_step50_noether_hinge.py`	3	R^2 0.92–0.99 vs ≤ 0.012; cert 10^{-16} vs 1.17
5	Lift to 3D contact	`experiments/step57_embodied_hinge.py`		3	clean-containment gate **INCONCLUSIVE**; Noether content lifts (invariant R^2=0.60–0.86 vs ≤ 0.06); resolved by Exp. 6
6	3D-aware containment	`experiments/step58_3d_containment.py`		3	E → ℓ=0 linear (R^2=0.62–0.91); L bilinear → ℓ=1 degree-2 cross (R^2=1.00, range 0.998–1.000)
7	Structure vs scale	`experiments/step51_structure_vs_scale.py`		3	equiv flat 1.1–1.2; in-wedge gain 31–166×; best baseline 10–155× above floor out-of-wedge
8	Approximate symmetry	`experiments/step53_approximate_symmetry.py`		3	cert exact at β=0 (68–320×); graceful ∝ ϵ (corr 0.88–0.98); threshold ϵ ≈ 0.01–0.06

Exp.	Result	Code	Test	Seeds	Headline number
9	**Certificate on real contact dynamics (PushT)**	`experiments/step59_pusht_certificate.py`		3	learned equivariant exactly flat over the orbit (ratio 1.00, equiv-resid $\sim 10^{-7}$) and competitive in-dist; no MLP scale (1.7k–272k) reaches the floor out-of-wedge (2.1–3.9×, $H=10$)
10	**Augmentation vs the certificate**	`experiments/step60_augmentation.py`		3	SO(2)-aug flattens a single PushT orbit (ratio 0.93–1.02 vs plain 1.84–2.75); on $\mathbb{Z}_2^6$ augmentation reaches a $\sim 10^{-4}$ floor but never the certificate's exact $\sim 10^{-32}$ from 7 generators ($\sim 10^{28}\times$)

Exp.	Result	Code	Test	Seeds	Headline number
11	**Certificate at the task level (closed-loop PushT pose control)**	`experiments/step61_closed_loop_certificate.py`	`tests/test_step61.py`	3	equivariant model + G-equivariant planner: closed-loop pose error orbit-invariant to the float floor (model-rollout & real-env ratio 1.000, all seeds) vs MLP ×1.1–2.2 (rollout) / ×1.6–3.6 (real-env); in-wedge competitive (3.6–16° vs 8–18°). Auxiliary cost-drift > 0.3 met 1/3 (eq $\sim 10^{-7}$ vs MLP 0.19–0.31)

Exp.	Result	Code	Test	Seeds	Headline number
12	**Certificate on** SO(3) **(3D point clouds, constructed teacher)**	`experiments/step63_se3_certificate.py`	`tests/test_se3_equivariance.py`	3	learned equivariant model exactly flat over the non-abelian SO(3) orbit (H=5 ratio 1.000, resid $\sim 10^{-5}$); MLP climbs ×2.1–5.7 out of the wedge. Structure-vs-scale in 3D: 7.4×-smaller equivariant carries the certificate; bigger MLP interpolates better in-wedge (0.19–0.27 vs 0.55–0.58). Honest: `compete` gate `INCONCLUSIVE` (equivariant accuracy floor high at laptop capacity — "flat is not good")

Exp.	Result	Code	Test	Seeds	Headline number
13	**Certificate on rendered pixels (C_4, frame averaging)**	`experiments/step64_frame_averaged_pixel.py`	`tests/test_step62.py`	3	frame-averaged pixel model orbit-flat to the float floor (ratio 1.000, equiv-resid $\sim 10^{-7}$); collapse-robust FVU matches/beats the unconstrained CNN and beats the steerable incumbent; rollout horizon-stable while steerable diverges. Honest: absolute FVU > 1 for *all* models (architecture-agnostic JEPA-latent property, not an equivariance cost)
14	**Horizon law on a learned model of real chaotic dynamics (Lorenz)**	`experiments/step70_lorenz_horizon.py`	`tests/test_step70.py`	3	learned one-step MLP of the Lorenz Δt-map (relMSE $< 10^{-4}$); certified-horizon staircase linear in $\log(1/\epsilon)$ (R^2=0.975–0.995) and the model's Lyapunov exponent (slope) *matches* the true λ_1=0.9056 to 1–8% ($\hat{\lambda}_1$=0.895/0.919/0.977). Prop. 7(a). The near-neutral PushT interior is the degenerate branch (b), R^2=0.02

Exp.	Result	Code	Test	Seeds	Headline number
15	**Horizon law across a class of chaotic systems**	`experiments/step71_multichaos_horizon.py`	`tests/test_step71.py`	3	same learned-model staircase on a 2D map + two flows: Hénon $\hat{\lambda}_1$=0.45–0.47 vs 0.419 (rel-err 8–12%, 3/3 seeds), Lorenz 1–5% (3/3), small-exponent Rössler 0.065–0.066 vs 0.0714 (8–9%, 2/3 — one seed's ~1500-step horizon under-crossed). Validates Prop. 8: the $O(\delta)$ bias falls 44% → 8% as fidelity rises (Rössler); the true-system staircase isolates the finite-T truncation (~9–10%)